Q MATHEMATICS 1

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2 FUNCTIONS OF ONE VARIABLE, PART 2

In this chapter we continue the investigation of some specific classes of functions.

In Section 1, we start with *broken* functions. As an example, we will build a model for the inventory management of a car dealer. The dealer has to decide, how many cars to order each time when his warehouse runs out. The larger his order size, the fewer orders are necessary; this will reduce the ordering costs, but the larger average inventory level will increase the holding costs. Correctly modeling *total* costs leads to a broken function.

Section 2 covers *power* functions. Our example here revolves around constructing a triangular gutter using a plate of length 100 and width 18. We want to bend the plate in such a way that the capacity of the gutter is maximal. Modeling this problem involves a root function, which is an example of a power function.

In Section 3, *exponential* and *logarithmic* functions are addressed. These functions play a crucial role in modeling processes of exponential growth and decay.

1 BROKEN FUNCTIONS

In this section we consider the class of functions which can be written as the *quotient* of two functions. Note that the domain of such a *broken* function is (apart from other possible considerations) restricted by the requirement that zeros are not allowed for the function that appears in the denominator.

EXAMPLE 1 A FENCE

In Example 5 of the previous chapter we considered the function given by the equation

$$L = 2\ell + \frac{200}{\ell}$$

which can be rewritten as

$$L = \frac{2\ell^2 + 200}{\ell}.$$

Here, ℓ denotes the length (in meters) of a piece of land with an area of 100 m^2, while L denotes the total length (in meters) of the fence that is used to encapsulate that piece of land. The challenge is to choose ℓ in such a way that the total length of the fence L is as

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small as possible. The rewritten version of the function above is an example of a broken function. After all, its numerator contains a quadratic function of ℓ, and its denominator a linear function. Obviously, the domain of the function is the interval $(0, \infty)$, i.e. with zero excluded. ◀

EXAMPLE 2 A CAR DEALER

A Toyota dealer sells 72 Toyotas of the type Auris per year. Each car has a value of 25 000 euros. Of course, the dealer's stocking policy involves costs. In order to analyze these costs, we start from the following assumptions:

- the sale of cars takes place *uniformly* over time
- every time when the cars kept in stock have been sold, the dealer *immediately* orders a number of new Toyotas, which can be supplied on the same day
- each time, the dealer orders the *same number* of cars.

We can distinguish two different kinds of costs here: *ordering costs* and *holding costs*.

- The **ordering costs** reflect the costs of transportation and administrative handling: they equal 250 euros per order, irrespective of the size of the order.
- The **holding costs** occur because cars in inventory require storage space and involve interest loss: on an annual basis, these costs are 4% of the total value of the cars in stock.

Of course the dealer wants to know the number of cars he should order each time, such that the *total* annual costs are minimal.

Before modeling this decision problem, we examine a special case: let's assume that the dealer has decided to order 6 cars each time.

■ **EXERCISE 1** (a) Find the annual ordering costs corresponding with this order size.

Due to our assumptions, the number of cars in the showroom will decrease *linearly*. For the situation where the dealer decides to order 6 cars each time, Figure 1 sketches the number of cars in the showroom, against time in months.

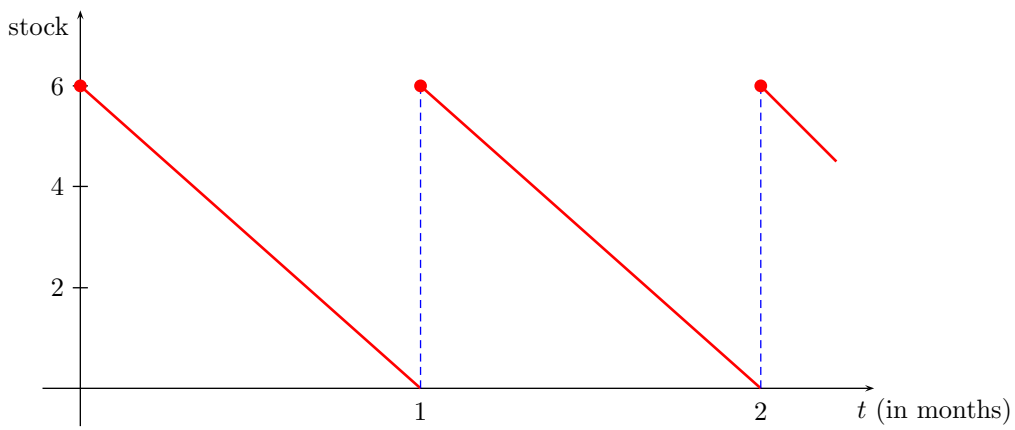

FIGURE 1 The stock level over time.

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■ **EXERCISE 1** (b) What is the average number of cars in stock?

(c) Find the annual holding costs.

(d) Find the total annual costs.

Let's now generalize our numerical example. To solve the car dealer's decision problem, we will need to determine the total annual costs for an *arbitrary* order size q.

■ **EXERCISE 2** (a) Find the number of orders per year, in terms of q.

(b) Find the annual ordering costs, in terms of q.

(c) What is the average number of cars in stock, in terms of q?

(d) Find the annual holding costs, in terms of q.

(e) Find the total annual costs, in terms of q. Show that your result can be rewritten in such a way, that the total annual costs are a broken function of q.

We trust that the previous exercise led you to the conclusion, that the total annual costs K are given by

$$K = \frac{18\,000 + 500q^2}{q}.$$

This formula defines a *broken* cost function k: $K = k(q)$. Figure 2 below sketches the graph of this function.

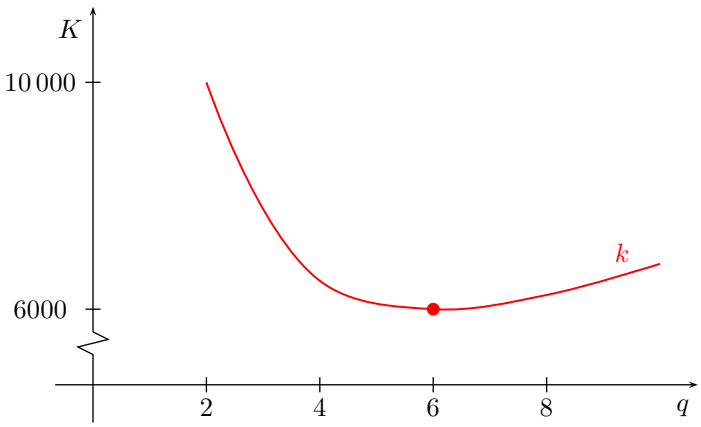

FIGURE 2 The total annual costs

Our challenge is to determine the order size q which minimizes costs. In Chapter 5 we will use the derivative of the function k to show that the car dealer should order 6 cars each month. ◀

■ **EXERCISE 3** In order to store his latest models, a Toyota dealer wants to build a new showroom. This showroom will be rectangular and its area should be 500 m^2. Three of its walls are to be made of reinforced concrete, costing 600 euros per meter. The fourth wall will be made of thick glass which costs 1200 euros per meter.

Let ℓ denote the length of the glass wall and let w be the width of the rectangle.

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(a) Express the total costs of building this showroom in terms of ℓ and w.
(b) Express w in terms of ℓ.
(c) Write the total costs of building this showroom in terms of ℓ.

2 POWER FUNCTIONS

A function f of the form

$$f(x) = x^a,$$

where a is some real number, is called a *power function*. Note that a quadratic function like

$$q(x) = 2 - x + 5x^2$$

is a 'combination' of the power functions $1, x$ and x^2 (corresponding with $a = 0, a = 1$ and $a = 2$, respectively).
Observe that the functions g and h defined by $g(x) = \sqrt{x}$ and $h(x) = \frac{1}{x}$, respectively, can also be written as a power of x. After all, the function g results if we take $a = \frac{1}{2}$:

$$\sqrt{x} = x^{\frac{1}{2}} \qquad (x \geq 0),$$

while the function h results if we take $a = -1$:

$$\frac{1}{x} = x^{-1} \qquad (x \neq 0).$$

As these two examples show, we must be a bit careful with respect to the domain of this type of functions.

EXAMPLE 3 POLYNOMIAL FUNCTIONS

By combining the power functions $1, x, x^2, x^3 \ldots$, we obtain so-called *polynomial functions*. An example of such a function is

$$p(x) = 3 + 2x - 5x^3 + 8x^4.$$

As x^4 is the highest power in this function, we say that the function p has *degree* 4. So a quadratic function is a polynomial function of degree two, and a linear function is a polynomial function of degree one. ◀

EXAMPLE 4 A DEMAND FUNCTION

A widely used specification for demand in empirical work is

$$D = A\,P^{-\alpha},$$

where P is the price. Further, A and α are positive constants. In Example 10 of the next chapter we will show that the demand function given by this equation has an appealing property. ◀

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EXAMPLE 5 A CONSTRUCTION PROBLEM

Using a plate of length 100 and width 18, we want to construct a triangular gutter as presented below. The idea is to bend the plate through the middle, lengthwise, in such a way that the capacity of the gutter is maximal.

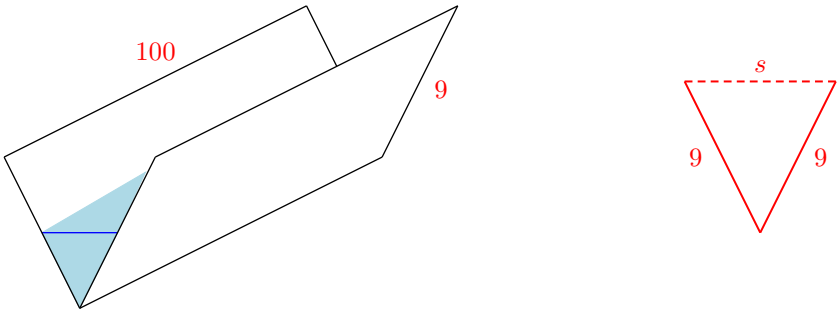

FIGURE 3 Constructing a gutter with maximal capacity

The capacity of the gutter, say C, is the area of the triangle presented in Figure 3, say A, multiplied by the length of the plate. So: $C = 100A$. In Example 4 of the previous chapter, we have seen that the area A of this triangle equals

$$A = \tfrac{1}{2}s\sqrt{81 - \tfrac{1}{4}s^2},$$

where s is the length of the third side of the triangle. Hence, the capacity of the gutter is given by

$$C = 50s\sqrt{81 - \tfrac{1}{4}s^2}.$$

Of course, the capacity is a function of the variable s, say $C = c(s)$. The function c turns out to involve a *root* function. Apart from the condition $s > 0$, the domain of the function c is determined by the requirement that the quadratic expression under the root sign must be nonnegative:

$$81 - \tfrac{1}{4}s^2 \geq 0 \Longrightarrow s^2 \leq 324 \Longrightarrow s \leq 18.$$

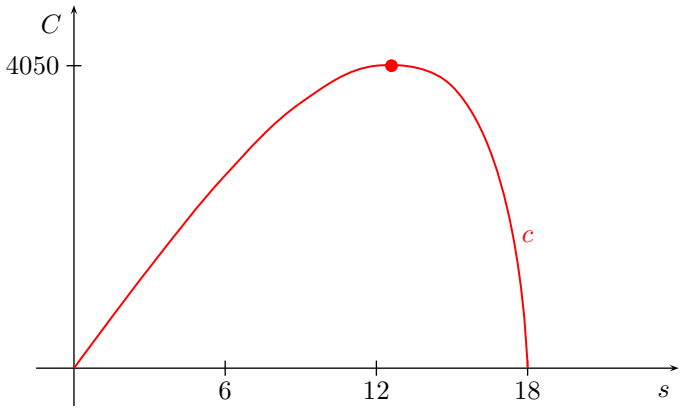

FIGURE 4 The capacity of the gutter

Figure 4 above sketches the graph of the function c. In Chapter 5 we will show that the capacity of the gutter is maximal for $s \approx 12.73$. ◂

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■ **EXERCISE 4** A restaurant has a main establishment on the beach, situated 100 meters from the sea at location H, and a playful sea restaurant built on pillars, which lies 200 meters into the sea at location Z. You can find a map picturing the situation, with the relevant distances, in Figure 5 below.

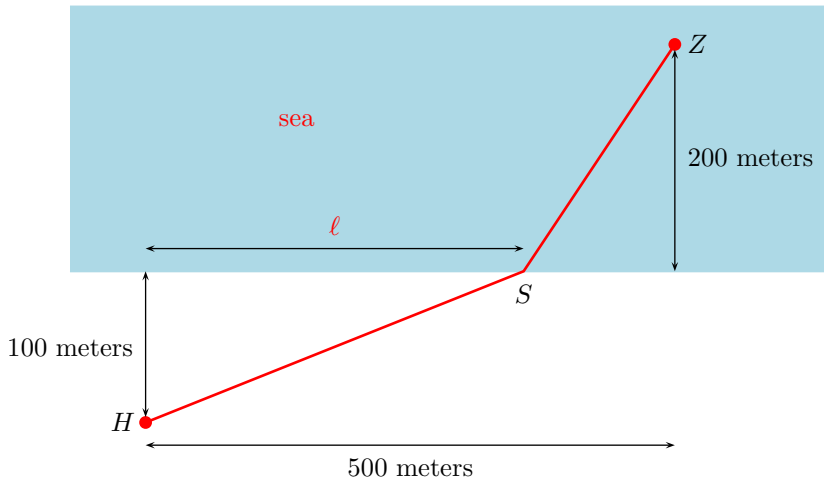

FIGURE 5 A map of the restaurant problem

All foods and drinks are prepared in the main establishment at H, and then have to be transported to the sea restaurant at Z. Transport takes place via the successive paths HS (using electrical scooters) and SZ (using electrical boats); the transport speeds over land and over sea are 200 and 100 meters per minute, respectively.

As you can see, Figure 5 implicitly defines a variable ℓ.

(a) Express the transport time T_1 from H to S in minutes in terms of the variable ℓ.

(b) Express the transport time T_2 from S to Z in minutes in terms of the variable ℓ.

(c) Express the total transport time T from H to Z in minutes in terms of the variable ℓ.

3 EXPONENTIAL AND LOGARITHMIC FUNCTIONS

In economics, but also in natural sciences like chemistry, many processes involve quantities that increase or decrease by amounts which are proportional to their size. In this context we speak of exponential growth (or exponential decay). We will demonstrate that such processes can be described by means of *exponential functions*. If we want to find out when this exponential growth (or decay) has reached a certain level, we will need *logarithmic functions*. We will illustrate most concepts on the basis of the following example of an exponential growth process.

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EXAMPLE 6 THE GROWTH OF CHINA'S GDP PER HEAD

In 2017, China's GDP (gross domestic product) per head, adjusted for purchasing power, amounted to around 14 000 euros. For the coming ten years, the Chinese planning agency predicts a trend growth of 6 % annually. Given these data, several interesting questions present themselves, e.g.:

- If the annual growth is 6 %, then how big is the growth per quarter, or per month?
- Starting from 2017, after how many years will GDP per head have increased by 25 %?◂

We will find an answer to these questions in Sections 3.1 and 3.2, respectively.

3.1 EXPONENTIAL FUNCTIONS

the first question

Let's address the *first* of these two questions step by step. In doing so, we will encounter all kinds of concepts that are linked to exponential functions.

dealing with the problem

Of course, if GDP per head grows by 6 % annually, then the size of this GDP is multiplied by a factor 1.06 every year. So if we write GDP_t for the GDP after t years, starting in 2017 from $GDP_0 = 14$ (i.e. 14 thousand euros), then the following holds:

$$\begin{aligned} GDP_1 &= 14 \cdot (1.06) = 14.84 \\ GDP_2 &= GDP_1 \cdot (1.06) = 14 \cdot (1.06)^2 = 15.73 \\ &\vdots \\ GDP_t &= 14 \cdot (1.06)^t. \end{aligned}$$

growth factor

We say that the development of China's GDP per head is characterized by *exponential growth*, where the constant number 1.06 is called the annual *growth factor*. The expression

$$14 \cdot (1.06)^t,$$

which we used to describe the growth of GDP, defines an exponential function.
Of course, we are not restricted to natural numbers for t in this function: it also possible to fill in other positive numbers. E.g., the value of GDP after a quarter is obtained simply by filling in $t = 0.25$ in the expression, and the growth factor per quarter therefore equals (in 4 decimal places) $(1.06)^{0.25} = 1.0147$. Note that this is slightly less than 1.015: this is a consequence of the principle of *compounding* which you may remember from secondary school, and will be addressed explicitly in QM2. Verify for yourself that the growth factor per month (again in 4 decimal places) equals 1.0049. This answers our *first* question.

EXPONENTIAL FUNCTION

A function f of the form

$$f(t) = b \cdot a^t,$$

is called an *exponential function* with *base* a, where the constant $a \neq 1$ is positive. The constant b is the value of f for $t = 0$.

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In our GDP example, $b = 14$ is the initial amount at $t = 0$, and the base of $a = 1.06$ is the growth factor per year.

Generalizing our GDP example, whenever an amount is multiplied by a constant factor per time unit, say by a, then we speak of exponential growth or decay. That any such process can indeed be described by the exponential function $f(t) = b \cdot a^t$ is easily shown:

$$f(t+1) = b \cdot a^{t+1} = b \cdot a^t \cdot a^1 = a \cdot \left[b \cdot a^t\right] = af(t).$$

In other words:

the value of f at any time $t+1$ is a times the value of f at the previous time t

This is the defining characteristic of an exponential growth process.

In Figure 6 below, the curve that represents the exponential function, say $y = a^t$, has been sketched for two very distinct cases: $a > 1$ and $0 < a < 1$. In particular, the graph on the left represents an exponential growth process with growth factor $a = 2$, while the graph on the right does so for $a = 0.5$. In both cases, we have taken $b = 1$, so that both graphs pass through the point $(0, 1)$.

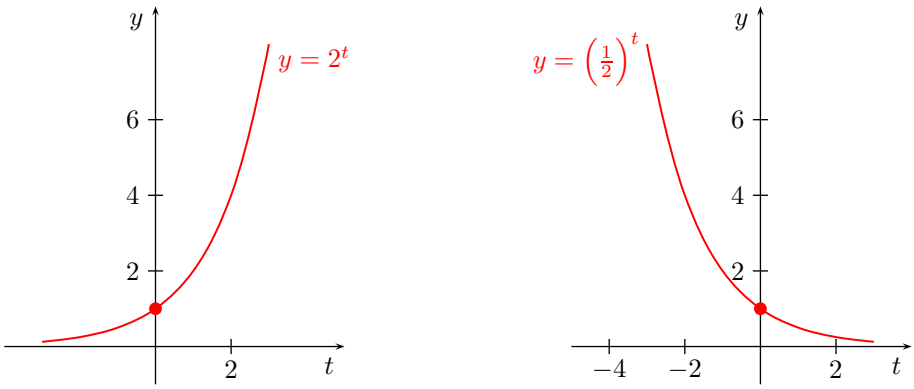

FIGURE 6 Two exponential functions

Note that the variable represented by the graph on the right is actually not *increasing* over time, but *decreasing*; its value is cut in half whenever a unit of time passes by. The following example describes a situation where such a process of *exponential decay* plays a role.

EXAMPLE 7 DEPRECIATION

If you own a car, you will be aware that its resale value drops every year. Let's assume that a car's value depreciates at an annual rate of 15 %. So if its value at $t = 0$ equals V_0, then its value after one year equals $V_1 = V_0 - 0.15V_0 = 0.85V_0$. Generalizing, after t years its value is $V_t = V_0 \cdot (0.85)^t$. For instance, $V_4 = V_0 \cdot (0.85)^4 \approx 0.52V_0$: so after 4 years the car's value has dropped to little more than half of its original value. ◀

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■ **EXERCISE 5** Find the exponential function for which the graph contains the following points:

(a) $(0, 2)$ and $(3, 16)$ (b) $(0, 3)$ and $(3, -\frac{1}{9})$.

For your convenience, we summarize some secondary school rules for manipulating exponential functions.

PROPERTIES OF EXPONENTS

If $a > 0$, $b > 0$ and x and y are numbers, then

(1) $a^0 = 1$ (2) $a^x \cdot a^y = a^{x+y}$ (3) $a^{-x} = \dfrac{1}{a^x}$

(4) $\dfrac{a^x}{a^y} = a^{x-y}$ (5) $\left(a^x\right)^y = a^{xy}$ (6) $(ab)^x = a^x \cdot b^x$

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the second question

We now continue our discussion of the two questions formulated in Example 6. Our *second* question was: starting from 2017, when will Chinese GDP have increased by 25 %, i.e. to $1.25 \times 14 = 17.5$? Let's start by sketching the relation between GDP and t in a graph.

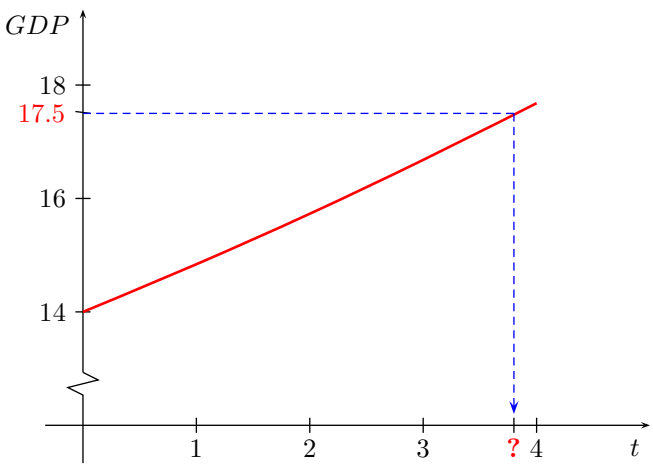

FIGURE 7 The growth of China's GDP per head

Remembering Section 1.4 from Chapter 1, we hope that many students immediately smell the notion of an 'inverse' here. After all, Figure 7 above shows how a value for the input variable t translates into a value for the output variable GDP. But our second question turns things around: it starts from a value of 17.5 for GDP as input, and we are looking for the implied value of t as our output. We'll formalize this issue in the next section.

3.2 LOGARITHMIC FUNCTIONS

exponential equation

To answer the second question from Example 6, we must solve t from the following so-called *exponential equation*:

$$14 \cdot (1.06)^t = 17.5 \Longleftrightarrow (1.06)^t = 1.25.$$

In words: we are looking for the value of t for which the exponential function with base 1.06 equals 1.25.

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Before presenting the numerical answer to this particular question, consider the more general exponential equation

$$a^x = c,$$

where $c > 0$. Two examples of such an exponential equation are $2^x = 8$ and $2^x = 6$. The first equation is easy to solve: since $2^3 = 8$, it is clear that $x = 3$ solves the equation. Finding the solution of the second equation is not that straightforward. Figure 8 illustrates the challenge.

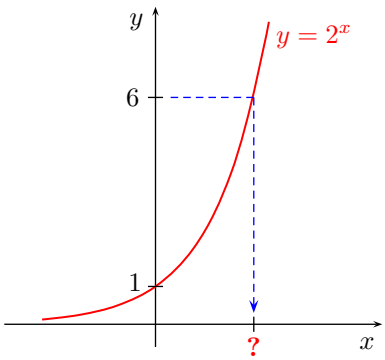

FIGURE 8 Solving the equation $2^x = 6$

To deal with an equation like $2^x = 6$, we need *logarithms*. In fact, we will denote the solution of this equation by

$${}^2\log 6$$

pronounced as 'two-log-six'. Note: 'two-log-six' is the exponent to which we must raise 2, in order to get 6. Consequently:

$$2^{{}^2\log 6} = 6$$

The number 2 is called the *base* of our logarithm. Using this 'language', the solution of the first equation, $2^x = 8$, can be written as ${}^2\log 8$. Apparently, ${}^2\log 8 = 3$. But how do we calculate the other logarithm, ${}^2\log 6$? Luckily, your pocket calculator has a '*log key*' for calculating logarithms with base 10. This may seem useless, because 'our' logarithm has base 2. But now the following formula (which we will not prove) is helpful:

$${}^2\log 6 = \frac{{}^{10}\log 6}{{}^{10}\log 2}.$$

Please use your calculator to verify that the result is about 2.58. Alternatively, you can check that $2^{2.58} \approx 6$. The rule below summarizes our result.

SOLVING AN EXPONENTIAL EQUATION

The solution of the exponential equation

$$a^t = c,$$

with $a > 0$, is given by

$${}^a\log c = \frac{\log c}{\log a},$$

where we follow the usual conventions and write $\log c$ instead of ${}^{10}\log c$.

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Now back to our original question: when will Chinese GDP have increased by 25 %, if the annual growth is 6 %? The implied exponential equation of $(1.06)^t = 1.25$ has the following solution:

$$^{1.06}\log 1.25 = \frac{\log 1.25}{\log 1.06} \overset{\text{(please verify!)}}{=} \frac{0.096910}{0.025305} = 3.83 \quad \text{(in 2 decimals).}$$

logarithmic function

So far, we have presented the logarithm as just a tool to solve exponential equations. Next, we will consider the so-called *logarithmic function* g, defined by

$$g(x) = {}^{a}\log x \qquad (x > 0),$$

where $a \neq 1$ is a positive real number. As an example, consider the case where $a = 2$. Like the left-hand side of Figure 6, Figure 9 below contains the graph corresponding to the function f defined by $f(x) = 2^x$. The careful reader has definitely recognized, that the *number* $^{2}\log 6$ is in fact equal to the value $f^{-1}(6)$. Generalizing, the *function* g defined by $g(x) = {}^{2}\log x$ is the *inverse* of the function f. Now remember from Section 1.4 of the previous chapter that the graphs of a function and its inverse are symmetric about the line $y = x$. Consequently, mirroring the graph of the exponential function with base 2 around the 45-degree line, we obtain the graph of the logarithmic function with base 2. Figure 9 illustrates the idea. Note in particular that, while the graph of the exponential function contains the point $(x, y) = (0, 1)$, its logarithmic inverse contains the point $(x, y) = (1, 0)$.

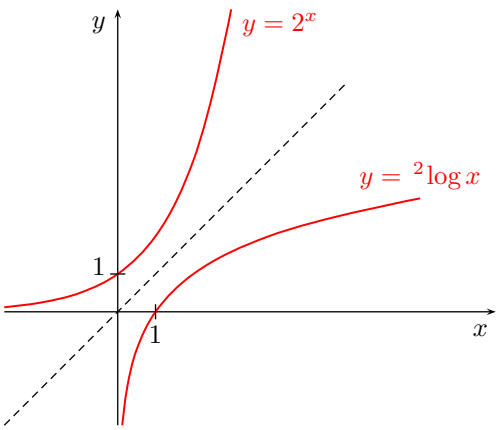

FIGURE 9 The graph of the logarithmic function with base 2

For your convenience, we summarize some secondary school rules for manipulating logarithmic functions.

PROPERTIES OF LOGARITHMS

If $a > 0, a \neq 1$ and x and y are positive numbers, then

(1) ${}^{a}\log 1 = 0$ (2) ${}^{a}\log(xy) = {}^{a}\log x + {}^{a}\log y$ (3) ${}^{a}\log\left(\frac{1}{x}\right) = -{}^{a}\log x$

(4) ${}^{a}\log\left(\frac{x}{y}\right) = {}^{a}\log x - {}^{a}\log y$ (5) ${}^{a}\log\left(x^y\right) = y\ {}^{a}\log x$ (6) ${}^{a}\log a = 1$

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In the next two examples, we will show how these properties can be helpful to solve exponential equations, or to simplify expressions involving logarithms.

EXAMPLE 8 To solve the equation

$$2^x = 5^{x+1}$$

we can start by taking the logarithm of both sides. We then apply properties (5) and (4), to obtain the following result:

$$\log 2^x = \log 5^{x+1} \iff x \log 2 = (x+1)\log 5 \iff x\big[\log 2 - \log 5\big] = \log 5$$
$$\iff x = \frac{\log 5}{\log 2 - \log 5} = \frac{\log 5}{\log(2/5)}.$$

Using your pocket calculator you will find that $x \approx -1.756$. ◀

EXAMPLE 9 The expression

$$\log 5 + \log 12 - \log 6$$

can be simplified considerably. Cleverly using properties (2), (4) and (6), we get

$$\log 5 + \log 12 - \log 6 = \log \frac{5 \times 12}{6} = \log 10 = 1.$$

Likewise, we can simplify the expression

$$10^{{}^{100}\log 9}.$$

It's easy to see that

$${}^{100}\log 9 = \frac{\log 9}{\log 100} = \frac{\log 9}{\log 10^2} = \frac{\log 9}{2\log 10} = \tfrac{1}{2}\log 9.$$

Hence, we can now develop our expression as follows:

$$10^{{}^{100}\log 9} = 10^{\frac{1}{2}\log 9} = \Big[10^{\log 9}\Big]^{1/2} = 9^{\frac{1}{2}} = \sqrt{9} = 3.$$ ◀

■ **EXERCISE 6** Express the following numbers as a multiple of $\log 5$.

(a) $\log 25$ (b) $\log \frac{1}{625}$ (c) ${}^{3}\log 25$ (d) $\log \sqrt{5} - \log \frac{1}{25}$

■ **EXERCISE 7** Solve the following equations.

(a) $2^x \cdot 3^x = 7$ (b) $2^{x^2-1} = 8$ (c) $\log x + \log x^2 = 9$ (d) $\log x = {}^{2}\log 12$

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END-OF-CHAPTER EXERCISES

EXERCISE 1 Match each of the graphs $A - F$ with one of the functions (a)-(e). In (f), try to find a suitable function which has the remaining graph.

(a) $y = \frac{1}{2}x^2 - x - \frac{3}{2}$

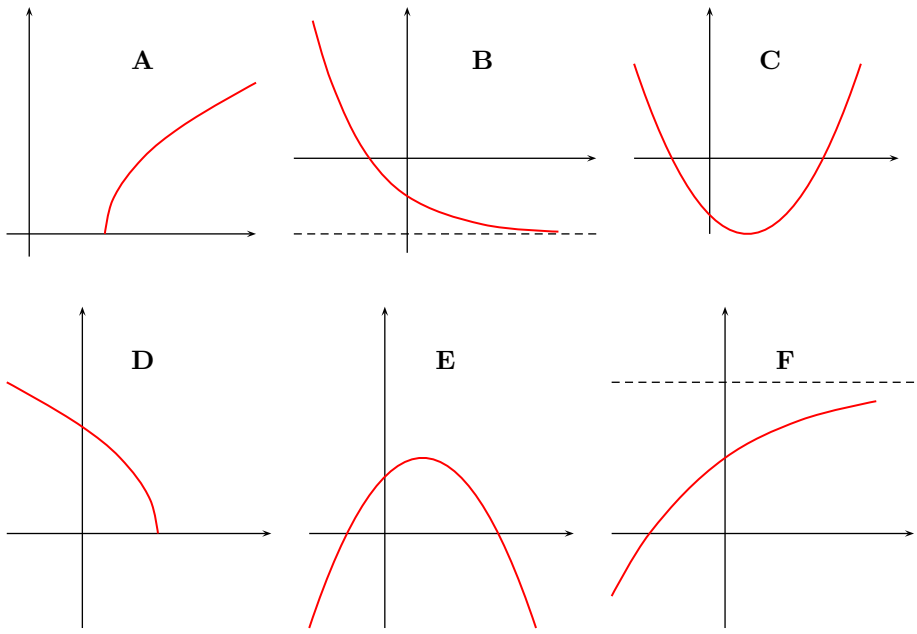
(b) $y = -\frac{1}{2}x^2 + x + \frac{3}{2}$

(c) $y = 2\sqrt{x-2}$

(d) $y = 2\sqrt{2-x}$

(e) $y = \left(\frac{1}{2}\right)^x - 2$

(f) $y = \cdots$.



EXERCISE 2 The volume V of a sphere depends on its radius r, according to the formula

$$V = \tfrac{4}{3}\pi r^3.$$

In other words V is a function of r: $V = g(r)$, where $g(r) = \frac{4}{3}\pi r^3$.

(a) Find the volume of a sphere with radius $r = 9$.

(b) What is the radius of a sphere with volume 27?

(c) Find the inverse function g^{-1}.

(d) Give a sketch of the graphs of the functions g and g^{-1}.

EXERCISE 3 The demand D for sugar in the U.S. in the period of $1915 - 1929$, as a function of the price P, was estimated by H. Schultz as

$$D = f(P) = \frac{157.8}{p^{0.3}}.$$

Here D and P are measured in appropriate units.

Solve the equation for P and so find the inverse of the function f.

EXERCISE 4 The population of Botswana was estimated to be 1.22 million in 1989, and to be growing at the rate of 3.4% annually. If $t = 0$ denotes 1989, find a formula for the population $P(t)$ at date t. What is the doubling time?

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EXERCISE 5 **Two growth processes**

(a) Let $f(t) = A \cdot 2^{rt}$ and $g(t) = B \cdot 2^{st}$, with $A > 0$, $B > 0$, and $r \neq s$. Solve the equation $f(t) = g(t)$.

(b) In 1990 the GNP (gross national product) of China was estimated to be $1.2 \cdot 10^{12}$, and the rate of growth was estimated to be $r = 0.09$. By comparison, the GNP for the USA was reported as $5.6 \cdot 10^{12}$, with an estimated rate of growth of $s = 0.02$. If the GNP of each country continued to grow exponentially at the rates $r = 0.09$ and $s = 0.02$, respectively, when would the GNP of the two nations be the same?

EXERCISE 6 In 1964, a five-year plan was introduced in Tanzania. One objective was to double the real per capita income over the next 15 years. What is the average annual rate of growth of real income per capita required to achieve this objective?

AN INSTALLATION PROBLEM

A water supply pipe is to be dug in from a residence V, which lies along an asphalted road, to a country cottage B, which lies in the middle of a forest. Installation of the pipe alongside the asphalted road costs 40 euros per meter; but installation through the moist forest soil costs 80 euros per meter. Below, you can see a map of the situation.

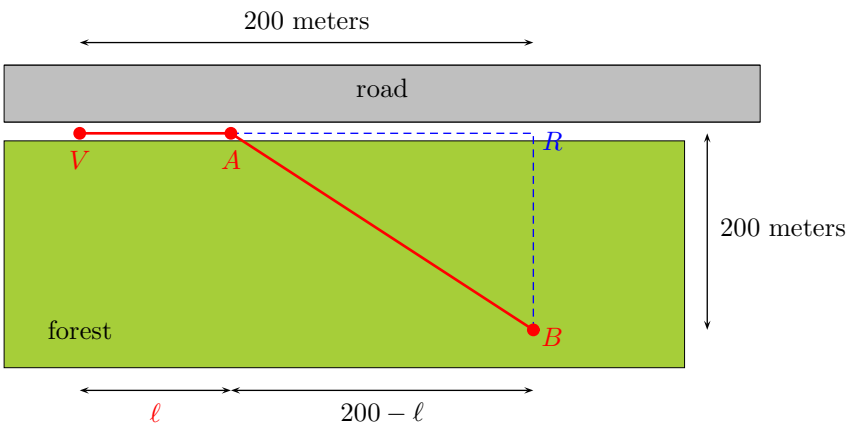

FIGURE 10 A map of the water pipe problem

To illustrate our options, the map does not only indicate the residence V and the cottage B, but also a point A somewhere along the asphalted road. The idea is that, starting from the residence, the pipe follows the asphalted road up to that point (path VA), after which we dig it in along a diagonal line through the forest towards the cottage (path AB).

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■ **EXERCISE 7** Assume that the pipe follows the asphalted road for the first 200 meters (path VR), then makes a 90 degree turn at R into the forest, to move straight to the cottage for another 200 meters (path RB).

What are the installation costs (in euros) of this option?

(a) Somewhere between 21 500 and 22 500 euros
(b) Somewhere between 22 500 and 23 500 euros
(c) Somewhere between 23 500 and 24 500 euros
(d) Somewhere between 24 500 and 25 500 euros.

■ **EXERCISE 8** Assume we install the pipe straight through the forest towards the cottage (path VB).

What are the installation costs (in euros) for this alternative?

(a) Somewhere between 21 500 and 22 500 euros
(b) Somewhere between 22 500 and 23 500 euros
(c) Somewhere between 23 500 and 24 500 euros
(d) Somewhere between 24 500 and 25 500 euros.

As you can see, the option described in Exercise 8 is cheaper than the alternative from Exercise 7. But of course, we have a continuum of intermediate alternatives here, where you follow the road *partly* (as depicted by the path VAB in Figure 10). To determine the installation costs for these intermediate cases, we define the variable ℓ as the distance from V to A.

■ **EXERCISE 9** The total installation costs K can be expressed in terms of ℓ by the following equation:

(a) $K = 40\ell + 8\sqrt{\ell^2 - 400\ell + 80\,000}$
(b) $K = 40\ell + 80(\ell^2 - 400\ell + 80\,000)$
(c) $K = 40\ell + 80\sqrt{\ell^2 - 400\ell + 40\,000}$
(d) None of the options (a), (b) or (c) is correct.

Of course, the installation costs are a function of the variable ℓ, say $K = k(\ell)$.

■ **EXERCISE 10** The domain of the cost function k is given by ...

(a) the interval [0, 100]
(b) the interval [0, 141.4]
(c) the interval [0, 200]
(d) None of the options (a), (b) or (c) is correct.

Hint: You can of course answer this question by formally investigating the function k, but also by showing some panache and drawing your conclusion directly from a critical look at Figure 10.

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SOLAR PANELS

An installation company specialized in solar panels offers its customers two options. The first option is that the company installs the panels, while the second option is to install them yourself. Under the *first option*, the company offers three different systems. The table below summarizes the most important information.

number of panels	6	10	14
price (in euro) of the system	3600	5600	7600
annual electricity yield (in kWh)	1500	2500	3500

Under the *second option*, a customer can order any number of panels: the costs are a *fixed* amount of 1500 euros, plus 300 euros per panel. The government, keen to promote the adoption of sustainable energy sources, offers a 10 % subsidy on the purchasing costs of solar panels, up to a maximum of 500 euros.

■ **EXERCISE 11** When you buy 10 panels, the subsidy amounts to ...

(a) less than 500 euro under option 1 and 500 euro under option 2
(b) less than 500 euro under option 1 and less than 500 euro under option 2
(c) 500 euro under option 1 and 500 euro under option 2
(d) 500 euro under option 1 and less than 500 euro under option 2.

A vital piece of information in the installation company's advertisements is the so called *payback period*. This is the number of years which it takes to earn back the purchasing cost of the panels through your savings on the cost of electricity. Let's assume that the price of electricity is, and continues to be, 22 eurocent per kWh.

■ **EXERCISE 12** When you buy 10 panels, the payback period is ...

(a) more than 9 years under option 1, and less than 7 years under option 2
(b) more than 9 years under option 1, and more than 7 years under option 2
(c) less than 9 years under option 1, and more than 7 years under option 2
(d) less than 9 years under option 1, and less than 7 years under option 2.

We now limit ourselves to *option two*.

■ **EXERCISE 13** The relation between the payback period T (in years) and the number of solar panels bought, say z (with $z \leq 11$), is given by ...

(a) $T = \dfrac{1350 + 270z}{55z}$
(b) $T = \dfrac{1500 + 300z}{55z}$
(c) $T = \dfrac{1350 + 270z}{250z}$
(d) None of the options at (a), (b) or (c) is correct.

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ENDANGERED SPECIES

Two species of animals are facing an acute extinction risk. Of the first species, only 2000 animals are left, and the population decays at a rate of 5 % per year. The second species is down to 1000 but, fortunately, this population is growing at a rate of 5 % per year.

■ **EXERCISE 14** By how many per cent does the population of the second species grow per *quarter*?

(a) Somewhere between 1.20 and 1.22 % per quarter
(b) Somewhere between 1.22 and 1.24 % per quarter
(c) Somewhere between 1.24 and 1.26 % per quarter
(d) Somewhere between 1.26 and 1.28 % per quarter.

■ **EXERCISE 15** After how many years from today will both populations be equally large?

(a) After between 4 and 5 years
(b) After between 5 and 6 years
(c) After between 6 and 7 years
(d) After between 7 and 8 years.

The environmental protection agency has decided to intervene when the population size of the first species falls below 1000. Such intervention could involve measures like the introduction of dedicated breeding programs in suitable zoos.

■ **EXERCISE 16** When will this intervention take place?

(a) After between 9 and 11 years from today
(b) After between 11 and 13 years from today
(c) After between 13 and 15 years from today
(d) After between 15 and 17 years from today.

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3 THE DERIVATIVE

This chapter introduces the concept of the *derivative*, and illustrates its use in economics and business. We hope you still remember from secondary school, that the derivative of a function measures something like 'the rate of change of that function at a certain point'. In Section 1, we will have a closer look at the basic idea.

In Section 2, we will present a few rules for determining the derivative (more precisely: the *derivative function*) for a limited class of basic functions. Additional rules will be presented next week, in Chapter 4.

For economists and business scientists, the notion of 'the rate of change of a function at a certain point' is of crucial importance. It allows them e.g. to determine, approximately but quickly, how profit or costs would change if sales volume changes by a small amount. Economists typically call such a small change a *marginal* change; by extension, the derivative function is then called the *marginal function*. We will illustrate all of this in Section 3, using among others the 'restaurant' example from Chapter 1.

In Section 4 we will study another, but related, core concept in economics and business: *elasticity*. Using the marginal function, *absolute* changes can be approximated, e.g. the change in sales as a result of a price change. But if we wish to answer the question by how many *percent* sales will change if price changes by e.g. 3 *percent*, then we are dealing with *relative* changes. This is where the elasticity concept comes in.

Although many of the issues covered in this chapter may be familiar to you, please do study it very carefully. As much as possible, and unlike in secondary school, the relevant mathematical concepts are treated from an economics and business perspective; the results constitute indispensable tools for later chapters and, without exaggeration, for almost all later *courses*.

1 THE IDEA BEHIND THE DERIVATIVE

The objective of this first section is to refresh your understanding of the concept of a 'derivative'. Our approach takes the study of rates of change as its starting point.

1.1 THE AVERAGE RATE OF CHANGE

In economics and business one often analyses the effects of changes. For example, if a firm considers to produce more, it wants to know how much its production costs will increase. We will start our discussion by introducing a way of measuring the rate at which one variable (say, production costs) changes in response to a change in the other (say, output level).

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■ **EXERCISE 1** Consider a company producing paint brushes. The daily costs (in euros) of producing q units (of 10 000 brushes each) are given by $K = q^3 - 12q^2 + 75q$. In other words: $K = k(q)$ with $k(q) = q^3 - 12q^2 + 75q$.

(a) Currently 80 000 brushes are produced per day, that is: $q = 8$.
Find the daily production costs corresponding with this production level.

(b) Now imagine that the output level q is increased by 2 units.
Find the daily production costs corresponding with this new production level.

(c) Find the increase in production costs.

(d) Find the increase in production costs **per unit of output**.

In the example below, we will generalize our findings from Exercise 1.

EXAMPLE 1 PRODUCING PAINT BRUSHES

Consider our paint brush producer from Exercise 1 again, with production costs given by $K = k(q)$. But unlike in Exercise 1, we now leave the cost function k *unspecified*. Figure 1 below, which has output q along the horizontal axis and costs K along the vertical axis, illustrates a possible shape for (a part of) the graph of the function k. The general shape of Figure 1 is consistent with the function which we considered in Exercise 1, but we emphasize that in the current analysis, k is an *arbitrary* function.

If, at the moment, 80 000 brushes are produced per day so that $q = 8$, then the corresponding production costs are $k(8)$ euros. In Figure 1, this starting scenario corresponds with the lower-left red dot. Now imagine that the output level is increased by Δq units, to $8+\Delta q$. At this higher output level, production costs are $k(8+\Delta q)$ euros. The new scenario corresponds with the upper-right red dot in Figure 1.

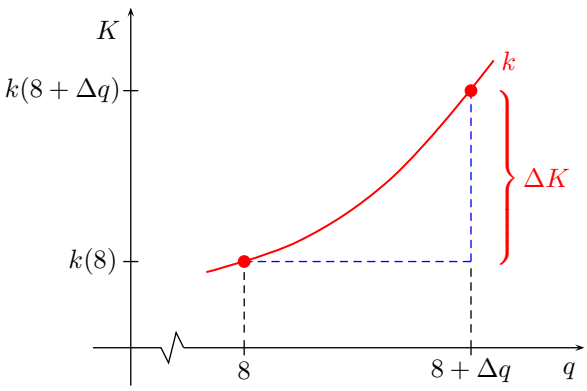

FIGURE 1 The change in production costs in response to an increase in production

Comparing both scenarios, the *change in production costs* measured in euros equals

$$\Delta K = k(8 + \Delta q) - k(8).$$

This change is indicated explicitly in Figure 1. Since production costs increase by ΔK if the firm increases its output by Δq, the increase in production costs **per unit of output** follows as

$$\frac{\Delta K}{\Delta q} = \frac{k(8 + \Delta q) - k(8)}{\Delta q}.$$

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average rate of change

This fraction, which is measured in euros *per unit*, is called the *average rate of change of k over the interval from* 8 *to* $8 + \Delta q$.

For the specific cost function used in Exercise 1, you should have found that the average rate of change of k over the interval $[8, 10]$ is

$$\frac{\Delta K}{\Delta q} = \frac{k(10) - k(8)}{10 - 8} = \frac{550 - 344}{2} = 103.$$

In words: over the interval from 8 to 10 units of output, the production costs are rising at an average rate of 103 euros per unit. ◀

■ **EXERCISE 2** Find, for the function k given in Exercise 1, the average rate of change over the interval $[8, 9]$. Relate your answer as explicitly as you can to Figure 1 above. In particular, try to motivate why your answer is *less* than the 103 euros per unit which we obtained over the interval $[8, 10]$.

After Example 1, it is easy to formally define the concept of 'the average rate of change' for an *arbitrary* function.

AVERAGE RATE OF CHANGE (or DIFFERENCE QUOTIENT)

Let f be a function of the variable x. The *average rate of change* of the function f over the interval $[a, a + \Delta x]$ equals

$$\frac{\Delta y}{\Delta x} = \frac{f(a + \Delta x) - f(a)}{\Delta x}$$

difference quotient

This fraction is also called the *difference quotient* of the function f corresponding with the change Δx.

Next, we discuss the crucial **geometrical interpretation** of our 'average rate of change'.

Figure 2 below repeats Figure 1, with a slight change in emphasis. Remember our two scenarios in Example 1: output q increases from 8 to $8 + \Delta q$, and as a result costs K increase from $k(8)$ to $k(8 + \Delta q)$. In Figure 2, these two scenarios correspond with the points P and Q.

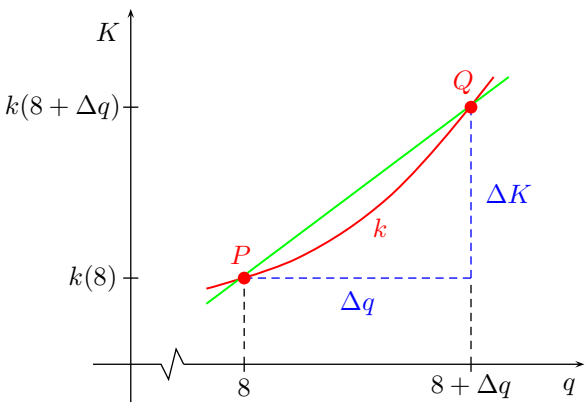

FIGURE 2 The change ΔK in K as a consequence of the change Δq in q

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From P to Q, the *horizontal displacement* equals the output rise of Δq; the *vertical displacement* equals the cost increase of $\Delta K = k(8+\Delta q) - k(8)$. Both these displacements are indicated explicitly in the figure. As discussed in Section 2.2 of Chapter 1, this implies that the *slope* of the (green) line through the points P and Q is equal to the quotient

$$\frac{\Delta K}{\Delta q} = \frac{k(8+\Delta q) - k(8)}{\Delta q}.$$

But that is precisely our 'average rate of change' formula! It will be clear that this observation generalizes beyond Example 1. Hence, for any *arbitrary* function f, we conclude that

the average rate of change of the function f over the interval $[a, a+\Delta x]$ is the slope of the line through the endpoints of the graph of f on this interval.

EXAMPLE 2 THE TEMPERATURE THROUGHOUT THE MORNING

Assume that, on some day, the temperature T (in degrees Celsius) between eight o'clock and twelve o'clock in the morning is given by the formula

$$T = 18 - \tfrac{1}{8}(t-16)^2,$$

where t is a number in the interval $[8, 12]$, representing time (in hours). Apparently, $T = f(t)$, with $f(t) = 18 - \frac{1}{8}(t-16)^2$. You can easily check that the temperature at eight and twelve o'clock equals, in turn, $f(8) = 10$ and $f(12) = 16$. This implies that the average rate of change of temperature over the time interval $[8, 12]$ is given by

$$\frac{\Delta T}{\Delta t} = \frac{f(12) - f(8)}{12 - 8} = \frac{16 - 10}{4} = 1.5.$$

So during the morning hours, the temperature is rising at an average rate of 1.5 degrees per hour. ◀

■ **EXERCISE 3** Find the average rate of change of temperature over the time interval $[8, 10]$.

1.2 THE RATE OF CHANGE

In the previous section we have learned how to calculate the *average* rate of change of a function over some *interval*. We will now discuss how a rate of change can be determined at a *single value*.

In Example 2 we found that the *average* rate of change of temperature over the time interval $[8, 12]$ is equal to 1.5 degrees per hour. But what can we say about the rate at which temperature is changing at exactly eight o'clock?

To find an answer to this question, let's start by determining the average rate of change over the time interval $[8, 8+\Delta t]$ for some values of Δt that come closer and closer to zero. By doing so, the endpoint $8+\Delta t$ of the interval comes ever closer to its starting value of 8. At Exercise 3 above, you have already seen that when Δt equals 2 rather than 4, the average rate of change turns out a bit higher than our original value of 1.5. In Table 1 below, we have included some even smaller values for Δt. Make sure you check the calculations!

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time interval with $\Delta t =$	4	1	0.5	0.1
average rate of change (in degrees per hour)	1.5	1.875	1.9375	1.9875

TABLE 1 The average rate of change of temperature over some 'shrinking' time intervals

It turns out that, starting from eight o'clock, the average rate of temperature change over shorter and shorter time intervals increases from 1.5 through 1.875 and 1.9375 to 1.9875. These numbers seems to be approaching the value 2. So we may suspect, that the rate of change (without 'average'!) **at** exactly eight o'clock is 2 degrees per hour.

Let's verify whether this suspicion is correct. In Table 1 we have been calculating the average rate of temperature change over some *specific* time intervals. If we leave the length Δt of the time interval *unspecified*, we find that the temperature change follows as

$$\begin{aligned}\Delta T &= f(8+\Delta t) - f(8) = 18 - \tfrac{1}{8}(8+\Delta t-16)^2 - 10 = 8 - \tfrac{1}{8}(\Delta t - 8)^2 \\ &= 8 - \tfrac{1}{8}\left[(\Delta t)^2 - 16\Delta t + 64\right] = 2\Delta t - \tfrac{1}{8}(\Delta t)^2.\end{aligned}$$

Consequently, the *average* rate of temperature change (i.e. the difference quotient) over the time interval $[8, 8+\Delta t]$ equals

$$\frac{\Delta T}{\Delta t} = \frac{2\Delta t - \frac{1}{8}(\Delta t)^2}{\Delta t} = 2 - \tfrac{1}{8}\Delta t.$$

Make sure to understand that all the scenarios in Table 1 are special cases of this equation: e.g., for $\Delta t = 1$, we get $2 - \frac{1}{8} = 1.875$. The question is now: what happens if Δt gets closer and closer and closer to zero, i.e. if we carry the sequence in Table 1 to its extreme? Clearly, the average rate of temperature change will then indeed get ever closer to 2. In mathematics, this observation is captured by the following formal notation:

$$\lim_{\Delta t \to 0} \left[2 - \tfrac{1}{8}\Delta t\right] = 2.$$

limit This is pronounced as follows: as Δt approaches 0, the *limit* of $2 - \frac{1}{8}\Delta t$ equals 2. Altogether, we have now shown that the *exact* value of the rate of change at eight o'clock equals 2. At that moment, temperature is rising at a rate of 2 degrees per hour.

Our example naturally leads us to the following definition of the rate of change for an *arbitrary* function.

RATE OF CHANGE (or DERIVATIVE)

Let f be a function of the variable x. The *rate of change* of the function f at $x = a$ equals

$$\boldsymbol{\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(a+\Delta x) - f(a)}{\Delta x}}$$

derivative This number will be denoted by $f'(a)$, and is usually called the *derivative* of the function f at $x = a$.

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Let's get back to our temperature example. As we have just seen, the rate of temperature change at eight o'clock, i.e. the derivative at that point, equals 2; so $f'(8) = 2$. Now what does that actually *mean*? Like so often, a geometrical representation is extremely helpful here. Figure 3 sketches our temperature function f, over an interval that includes $t = 8$ (where $T = 10$) and $t = 12$ (where $T = 16$).

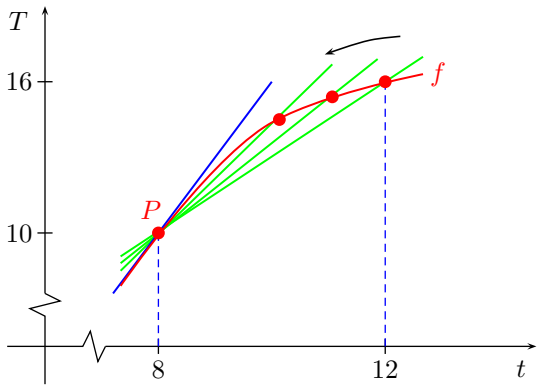

FIGURE 3 Choosing Δt closer and closer to zero

As we already saw in Example 2, when we start at $t = 8$ and take $\Delta t = 4$, the average rate of change over the interval $[8, 12]$ follows as $\frac{16-10}{12-8} = 1.5$. This value corresponds with the slope of the line that passes through the points $(8, 10)$ and $(12, 16)$. From the figure, it is easy to see that if we take smaller values for Δt, e.g. 3 or 2, the slopes of the implied green lines become larger. In fact, if we choose Δt closer and closer to zero, then the (slopes of the) respective green lines come closer and closer to the (slope of the) blue line that 'touches' the graph of f at the point P. This implies that the slope of that blue line corresponds with $f'(8)$; so in the case at hand, it has a value of 2. This blue line is called the *tangent* to the graph of f at the point P. Using the 'point-slope' formula from Section 2.2 in Chapter 1, we find that

$$T = 2(t - 8) + 10$$

is an equation of this tangent.

Like before, we generalize our result to *arbitrary* functions.

TANGENT

The line through the point $\big(a, f(a)\big)$ with slope $f'(a)$ is called the *tangent* to the graph of f at the point $\big(a, f(a)\big)$. An equation of this tangent is $y = f'(a)(x - a) + f(a)$.

EXAMPLE 3 Consider again the cost function k introduced in Exercise 1: $k(q) = q^3 - 12q^2 + 75q$. Let's determine the rate of change $k'(8)$ of this function at $q = 8$. Paraphrasing our analysis for the temperature function, we can proceed in three steps.

step 1 First, we determine the increase in production costs ΔK which results from an unspecified rise Δq of the production level (starting from the current level of $q = 8$). Removing the brackets and simplifying the resulting expression (check!) leads to:

$$\Delta K = k(8 + \Delta q) - k(8) = \big(\Delta q\big)^3 + 12\big(\Delta q\big)^2 + 75\Delta q.$$

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step 2 Next, we compute the difference quotient $\Delta K/\Delta q$:

$$\frac{\Delta K}{\Delta q} = (\Delta q)^2 + 12\Delta q + 75.$$

step 3 Finally, we evaluate the limit of this expression when Δq approaches 0:

$$k'(8) = \lim_{\Delta q \to 0} \frac{\Delta K}{\Delta q} = 75.$$

In other words: the rate of change of the cost function k at the production level $q = 8$ equals 75. At this production level, production costs are rising by 75 euro per 10 000 brushes. Geometrically, this number corresponds to the slope of the tangent to the graph of k at point P in Figure 2. An equation for that tangent is

$$K = 75(q - 8) + 344.$$

EXERCISE 4 Consider the function f given by $f(x) = x^2$.

(a) Find the difference quotient for the function f on the interval $[1, 1 + \Delta x]$.

(b) Determine $f'(1)$.

(c) Determine an equation for the tangent to the graph of f at the point $(1, 1)$.

At Exercise 4(b), we trust you managed to find that $f'(1) = 2$. Of course, since the function f is nonlinear, the value of e.g. $f'(2)$ or $f'(3)$ would be different. Generalizing, let's see what happens at an *arbitrary* point in the domain of f, say for $x = a$.

EXERCISE 5 Consider again the function f given by $f(x) = x^2$. Let a be some number.

(a) Find the difference quotient for the function f on the interval $[a, a + \Delta x]$.

(b) Show that $f'(a) = 2a$.

Typically, we use the symbol x, rather than a, to denote a variable that can take any value. Sticking to this convention, we can rewrite your result at Exercise 5(b) as

$$f'(x) = 2x.$$

This notation brings out that, by evaluating the derivative of our function, we have actually constructed a *new function*. This new function, denoted by f', is called the *derivative function* of f (or just 'the derivative' for short). At any point on the graph of f, the value of f' tells us the rate of change of f at that point. Equivalently: it tells us the slope of the tangent to the graph of f at that point.

JARGON ALERT. In many books and articles, alternative notations are used for the derivative. Instead of $f'(x)$, you will often encounter $\frac{\mathrm{d}f}{\mathrm{d}x}(x)$, or simply $\frac{\mathrm{d}y}{\mathrm{d}x}$.

Make sure you stay on track here. Until half a page ago, we have discussed how to find the value of the derivative of some function at a *specific* point. E.g. for the function in Exercise 4, the derivative at $x = 1$ turns out to be 2. We have now seen how to find an expression for the derivative of a function at *any* point. That expression is then in itself a function.

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E.g. the function $f'(x) = 2x$ tells us the derivative of f for any value of x. And of course, the derivative at $x = 1$ is then a special case.

Do all functions have a derivative function? No. Sometimes, for some points in the domain of a function, the value of the derivative cannot be obtained; strictly speaking, in such cases, the derivative function does not exist. In the next example, we show three typical manifestations of this phenomenon.

EXAMPLE 4 Consider the graphs of the functions a, j and r defined by

$$a(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \qquad j(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases} \qquad \text{and} \quad r(x) = \sqrt{x} \qquad (x \geq 0),$$

respectively.

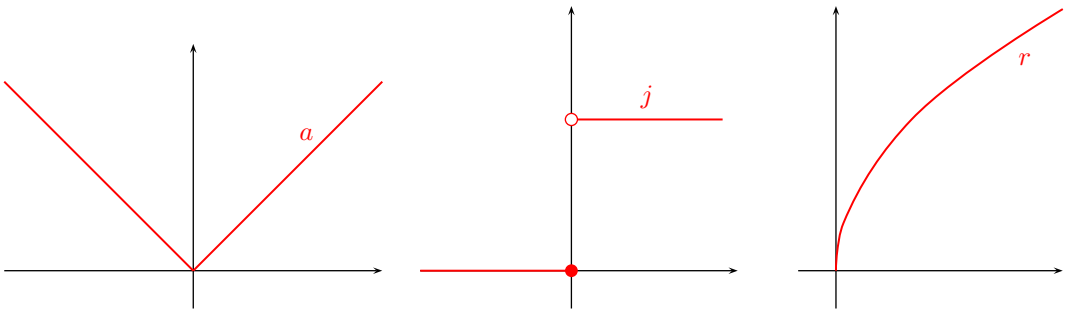

FIGURE 4 Three functions for which the derivative function does not exist

The graphs of all three functions contain the origin. And it is precisely at $x = 0$ that their derivatives do not exist. We could show this formally by analyzing the relevant difference quotients. Instead, let's have a careful look at the graphs.
Consider the graphs of the absolute value function a in the left panel, and the jump function j in the middle. It should be obvious that in both cases you can draw more than one line that 'touches' the graph of the function at the origin. So in both cases there is no unique tangent at the origin.
The graph of the root function r in the right-hand panel does have a unique tangent at the origin, but this tangent is *vertical*. Such a vertical line does not have a well-defined slope. Altogether, we conclude that none of these three functions has a derivative at $x = 0$. Consequently, the derivative functions do not exist; at least, not on the entire domain of the original functions. ◀

2 FINDING THE DERIVATIVE OF SOME BASIC FUNCTIONS

In this section, we will present a general expression for the derivative of an important class of basic functions defined in Chapter 2: *power functions*. Furthermore, we will see how to find the derivative of very simple combinations of basic functions. In next week's Chapter 4, we will address the derivatives of other classes of basic functions, and show how to deal with more complicated combinations of basic functions.

Up to now, we have used the difference quotient to find the derivative of a function, in particular in Exercise 5(b). Doing so was a rather tedious job! Now suppose that we want

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to calculate the derivative of a much more complex function: the profit function w from Section 2.3 in Chapter 1, defined by

$$w(p) = -4p^2 + 268p - 3640.$$

You should recognize from Section 2 in Chapter 2, that this quadratic function is an example of a *polynomial* function of degree 2. In other words, it is a 'combination' of basic power functions: in particular, the quadratic function p^2, the linear function p and the constant function 1. So if we knew the derivatives of these basic functions, and if we knew how to take the derivative of a *combination* of a number of functions, then finding the derivative function w' of our profit function would be a very simple exercise. We will now first discuss the relevant rules; having done so, we will get back to the derivative of our profit function at Example 5.

■ **EXERCISE 6** Use the difference quotient to show that the derivative of the constant function f defined by $f(x) = 1$ equals 0. How can you interpret your result geometrically, i.e. in terms of the graph of f?

The result at Exercise 6 generalizes: *the derivative of a constant function equals zero.*

Let's now have a look at the general class of *power functions*, i.e. functions of the form $f(x) = x^r$. It will be clear that basic constant, linear and quadratic functions belong to this class, in particular: for $r = 0$, $r = 1$ and $r = 2$, respectively. However, we repeat that the exponent r does not have to be a *natural* number; it can be any real number. E.g. the function $f(p) = \dfrac{1}{p} = p^{-1}$ is also a power function, with exponent $r = -1$.
The following rule allows us to find the derivative of a power function quickly.

THE DERIVATIVE OF A POWER FUNCTION

If f is a power function, i.e. $f(x) = x^r$, with r some real number, then $f'(x) = rx^{r-1}$.

Note that your results at Exercises 6 and 5(b) are actually special cases of the rule above, for $r = 0$ and $r = 2$. In fact, it would be possible to prove the rule by generalizing your analyses back there. We'll spare you going through all that. We do emphasize that, when we apply the rule above, we must be careful with the values of x which we choose. For instance, the root function $g(x) = \sqrt{x}$ is defined for *all* $x \geq 0$. By writing

$$g(x) = \sqrt{x} = x^{\frac{1}{2}},$$

we can see that g is a power function, with $r = \frac{1}{2}$. So its derivative function follows as

$$g'(x) = \tfrac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2x^{\frac{1}{2}}} = \frac{1}{2\sqrt{x}}.$$

However, this derivative function is only defined for $x > 0$, i.e. *not* for $x = 0$. In other words: the domain of the derivative function is a little bit smaller than the domain of the original function here. We already observed this complication, using a geometrical argument, in Example 4.

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Next, we come to the problem of finding the derivative of a *combination* of (basic) functions. For the moment, we will address two ways of combining functions:

- adding them up, and
- multiplying them with a constant.

For these combinations of functions the following two rules, which you probably remember from secondary school, are essential.

THE SUM RULE

If the derivatives f' and g' exist, then

$$\boldsymbol{\Big[f(x)+g(x)\Big]' = f'(x)+g'(x)}$$

and

THE SIMPLE PRODUCT RULE

If the derivative f' exists and c is some constant, then

$$\boldsymbol{\Big[cf(x)\Big]' = cf'(x)}$$

EXAMPLE 5 Applying the rules above, which you will probably do almost subconsciously, we find that the derivative of the profit function w equals

$$\Big[-4p^2+268p-3640\Big]' = \Big[-4p^2\Big]' + \big[268p\big]' - \big[3640\big]' = -8p+268.$$

Clearly, the standard rules above allow us to find this derivative in a much more convenient way than by applying the definition of the difference quotient.

EXAMPLE 6 In order to differentiate the function f defined by

$$f(x) = 3x\sqrt[3]{x} + \frac{2}{x^2} \qquad (x \neq 0),$$

we can start by rewriting the function more explicitly in terms of power functions:

$$f(x) = 3x \cdot x^{\frac{1}{3}} + 2x^{-2} = 3x^{\frac{4}{3}} + 2x^{-2}.$$

If we now apply the Sum Rule and the Simple Product Rule, it follows easily that

$$\Big[3x^{\frac{4}{3}} + 2x^{-2}\Big]' = 3 \cdot \tfrac{4}{3}x^{\frac{1}{3}} + 2 \cdot (-2)x^{-3} = 4\sqrt[3]{x} - \frac{4}{x^3}.$$

EXERCISE 7 Determine the derivative of the following functions:

(a) $f(x) = 2x^2 + \dfrac{4}{x^2} + 15 \quad (x \neq 0)$ (b) $f(t) = t\sqrt{t} + 500t \quad (t > 0)$

(c) $f(x) = \sqrt{2x} \quad (x > 0)$ (d) $f(t) = (t-1)(t+1)$.

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■ **EXERCISE 8 RATE OF PRODUCTION**

An assembly plant has an 8-hour working day. An analysis of the production data has shown that, on a typical day, the number of finished units produced **after** t hours of work can be described very well by means of the expression

$$q(t) = 50t + 2t^2 + \tfrac{1}{8}t^3 \qquad (0 \le t \le 8).$$

Note that $q(0) = 0$, while $q(8) = 592$. In other words: at the start of a working day, there are no finished units of output yet, but as the day proceeds the number gradually increases and by the time the workers go home, 592 units are ready for shipment.

(a) Find the number of units produced during the period $[2, 4]$.

(b) Find the *average* number of units produced per hour during that period.

(c) Determine $q'(2)$ and give an interpretation of this number.

3 MARGINAL ANALYSIS

Remember that, in Section 1, we have *defined* the concept of a derivative; next, in Section 2, we saw how to *determine* that derivative for a limited class of functions. In the current section and the next, we will address two crucial *applications* of the derivative in economics and business: *marginal analysis* and *elasticity*.

In general terms, marginal analysis refers to the study of the change of an economic quantity, in response to a small change in one of its determinants. Two examples may illustrate why this is an important topic.

- A brand manager will obviously have a fair idea of how many units his product sells at the price which is currently being charged. But typically, he will also be interested to know how much demand would increase (or decrease) when he lowers (or raises) its price by a certain amount.
- A manufacturer is not only interested in the total costs corresponding to the current production level. Typically, he also wants to know how much these costs will change if the production level changes.

EXAMPLE 7 SENSITIVITY WITH RESPECT TO PRICE CHANGES

In Example 10 of Chapter 1, we discussed a restaurant manager who considers to increase the current price of 30 euros of the plat du jour. In this example a major role was played by the *linear* demand function d and the *quadratic* profit function w. Our analysis back then implied that, for a general price level of p euros, each week the menu is ordered by q customers and the profits are W (in euros), where

$$q = 220 - 4p$$

and

$$W = -4p^2 + 268p - 3640,$$

respectively. So $q = d(p)$, where d is the linear demand function as defined by the right-hand side of the first equation above and $W = w(p)$, where w is the quadratic profit function as defined by the right-hand side of the second equation above.

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Before the manager takes a decision, he wants to know how 'sensitive' demand and profits are with respect to changes in the price level. So let's assume that the price changes by an amount Δp (starting from the current price of 30 euros).

the customers

Then the number of customers changes by

$$\Delta q = d(30 + \Delta p) - d(30) = \big[220 - 4(30 + \Delta p)\big] - 100 = -4\,\Delta p.$$

So increasing the price by 25 cents will *decrease* the number of customers by $4 \times 0.25 = 2$. Since $d'(30) = -4$, we find that

$$\Delta q = d'(30)\,\Delta p$$

the profits

To find the effect on the weekly profits, we assume that Δp is *small*. Then the difference quotient

$$\frac{\Delta W}{\Delta p} = \frac{w(30 + \Delta p) - w(30)}{\Delta p}$$

is bound to be *close to* its limiting value, i.e. the derivative $w'(30)$. We can denote this as follows:

$$\frac{\Delta W}{\Delta p} \approx w'(30).$$

As a consequence,

$$\Delta W \approx w'(30)\,\Delta p$$

Since $w'(30) = 28$ (check!), increasing the price by 25 cents will *increase* the weekly profits by *approximately* $28 \times 0.25 = 7$ euros. Make sure to check that in this case the *exact* increase in profits, i.e. $w(30.25) - w(30)$, equals 6.75 euros. So our approximation is pretty close here. ◀

To carry our findings over to a more general setting, first consider an arbitrary (nonlinear) function f describing a relation between the variables x and y: $y = f(x)$. If the derivative of f exists at $x = a$, then, for a small change Δx (starting from a), we have

$$\frac{\Delta y}{\Delta x} = \frac{f(a + \Delta x) - f(a)}{\Delta x} \approx f'(a).$$

This implies that

$$\Delta y \approx f'(a)\,\Delta x.$$

We have now obtained the following important result.

Δ-FORMULA

If $y = f(x)$ and the derivative of the function f at $x = a$ exists, then

$$\boldsymbol{\Delta y \approx f'(a)\,\Delta x}$$

Here Δx is a *small* change in the variable x (w.r.t. the value a) and Δy is the resulting change in the variable y.

In words:

the derivative translates a small change in the independent variable into an approximation for the resulting change in the dependent variable.

Of course for a linear function ℓ we have $\Delta y = \ell'(a)\,\Delta x$: in other words, the Δ-formula will give the *exact* result.

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Summarized in a table:

	a linear function ℓ	*a nonlinear function* f
$x : a \to a + \Delta x$	$\Delta y = \ell'(a)\,\Delta x$	$\Delta y \approx f'(a)\,\Delta x$

TABLE 2 Functions of one variable: the response to a small change in x

Figure 5 provides a geometrical interpretation of the Δ-formula and, not surprisingly, the tangent plays a crucial role here. The issue at hand will be clear: we are interested in the *increase* Δy of the graph of the function f when we increase x from a to $a + \Delta x$. The left panel of the figure gives us the *exact* answer: Δy corresponds with the distance between the point P (on the graph of f) and the *horizontal* blue (dotted) line.
Geometrically, the idea behind our Δ-formula is

to approximate the actual graph of the function f by its tangent at $x = a$.

As we learned in Example 7, the *exact* increase of the tangent (which is the graph of a **linear** function!) equals the product of the slope of the tangent – which is $f'(a)$ – and Δx. The right panel of the figure shows that the the increase $f'(a)\,\Delta x$ of the tangent corresponds with the distance between the point Q (on the tangent to the graph of f) and the *horizontal* blue (dotted) line. As long as the points P and Q are close together (i.e. if Δx is small), these two distances will be similar, and the formula gives us an acceptable approximation.

Note that the Δ-formula gives the *exact* result in case of a linear function because for such a function its graph and the tangent *coincide*.

to get a clear picture our Δx is quite large

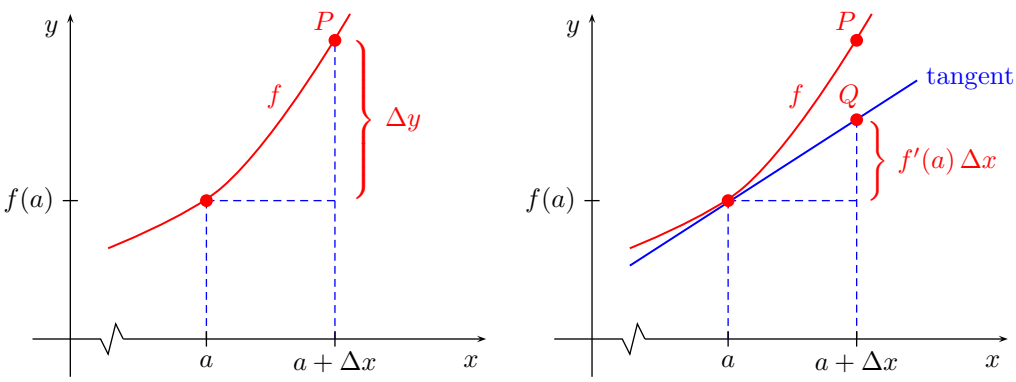

FIGURE 5 The (blue) tangent is close to the (red) graph.

■ **EXERCISE 9** Assume that the management of the restaurant discussed in Example 7 wants to increase its weekly profits by 24 euros.

(a) Find an *approximation* for the new price which would be needed to generate this profit increase.

(b) What is the *exact* new price that would be needed to generate this profit increase?

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EXAMPLE 8 MARGINALITY

In Example 7, we found that a change of the menu price by Δp (starting from an initial price level of 30) leads to a change in profits equal to

$$\Delta W \approx w'(30)\,\Delta p.$$

Economists often want to know by how much profit changes if the price rises by precisely **one** unit. In that case we have $\Delta p = 1$, so our approximation implies a profit rise of around $w'(30) = 28$.

Economists refer to $w'(30)$ as the *marginal profit* at $p = 30$. This concept of *marginality* is absolutely essential in economics: in later courses, you will constantly run into concepts such as marginal profit, marginal revenue, marginal cost, marginal utility, the marginal propensity to consume, etcetera. The basic idea is always the same: we use the derivative to approximate the change in some dependent variable (e.g. cost) in response to a one-unit change in some independent variable (e.g. sales volume). ◀

In the next exercise, you can apply what you have just learned. Be sure to pay close attention to the units in which production costs and production level are measured!

■ **EXERCISE 10** A company located on the Sand Algarve in Southern Portugal produces salt using salt pans. The company economist has used the weekly production data for the past year to estimate the following cost function:

$$k(q) = 0.001q^3 - 0.05q^2 + 1.2q + 12 \qquad (0 \le q \le 60).$$

Here q denotes the weekly salt production in tons (i.e. thousands of kilos), while the weekly production costs are given in thousands of euro.

(a) Starting from a weekly production level of 50 tons, use the concept of *marginal cost* (MC) to approximate the cost of producing 100 kilograms of salt extra. How good is this approximation?

(b) Like most firms, our salt producer faces demand fluctuations: in some weeks demand falls, in others it rises. The firm's economist is convinced that demand increases should *not* be met if marginal cost exceeds 0.4. To what interval is production restricted, if it is required that $MC \le 0.4$?

4 ELASTICITY

In Section 3, we have seen that the *marginal* or *derivative* of a function (e.g. a profit or a cost function) can be used to approximate the changes in the dependent variable, if the independent variable changes by a small amount. This approach is all about *absolute* changes: e.g. if, at some point, the derivative of a demand function equals -3.4, then a price increase by 1 unit will approximately lead to a decline in sales volume by 3.4 units. But often economists are more interested in *relative* changes: e.g., by how many percent will sales volume fall if the price is raised by 1%? The answer to this question is provided by the *elasticity* of a function. Like 'marginality', 'elasticity' is another core concept in economics and business, and (after the next exercise) we will illustrate it by means of an example.

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■ **EXERCISE 11 RELATIVE CHANGES**

Imagine that some price p is raised by 3 units. So the absolute price change is $\Delta p = 3$.

(a) Find the size of the relative price change if the initial price is $p = 15$.

(b) Find the relative price change in percentages if the initial price is $p = 15$.

(c) Write down an expression for the relative price change if the initial price p and the absolute price change Δp are unspecified.

EXAMPLE 9 A FERRY

Consider a ferry which car drivers can use to cross a river. Currently, the ferry-man charges a fare of 2.50 euro per car for a one-way crossing. Of course, the daily number of cars that make a crossing, say q, will depend on the fare p according to some demand function d, so $q = d(p)$. An economic consultant has estimated that the function d can be modeled by means of the following equation:

$$q = 1000(1 - \tfrac{1}{2}p + \tfrac{1}{16}p^2) \qquad (0 \leq p \leq 4).$$

The ferry-man considers raising his fare, e.g. to 2.75 euro per crossing, because he hopes to increase his income. In response, some car drivers will then inevitably stay away (e.g. to make a detour and use a bridge instead). The ferry-man understands that, if too many car drivers do so, his income will actually decrease rather than increase. So ultimately, he will *somehow* have to compare the changes in the related variables q and p. ◀

The ferry-man from Example 9 is clearly interested in the impact which a price change, say by 0.25 euro, will have on the demand for his services. How to measure that impact? An obvious first idea is to determine the difference

$$d(2.75) - d(2.50) = 97.65625 - 140.625 \approx -43.$$

In words: if we increase the price from 2.50 to 2.75 euro i.e. by 0.25 euro, then demand will fall by around 43 crossings. Undeniably, this number does give us some idea of how demand responds to changes in the price. But crucially, the above difference is an *absolute* change.

To get a clear idea of the sensitivity of the number of crossings with respect to the fare, it is probably better to think in terms of *relative* changes. After all, an absolute price increase of e.g. 0.25 euro can mean many things. A price increase from 1 to 1.25 euro (say: for a candy bar) constitutes a massive 25% rise, whereas the same absolute increase starting from 1000 euro (say: for a flatscreen TV) only means a rise of 0.025%; almost imperceptible! In the example at hand, with an original fare of 2.50 euro, those 25 cents correspond to a 10% price rise. Similarly, an absolute change in demand by 43 crossings doesn't mean anything in isolation; such a number can only be understood within its appropriate context, i.e. against the background of the number of crossings *before* the price rise. In the case of the ferry-man, the *relative* change of demand follows as:

$$\frac{d(2.75) - d(2.50)}{d(2.50)} = -\frac{43}{140.625} = -0.306$$

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or minus 30.6%. So we conclude that a 10% price rise will lead to a decline in demand by no less than 30.6%. That sounds like a massive response; something which would have been much less obvious from the original *absolute* change. In fact, your intuition should now suggest that raising the fare may not be such a good idea here.

Our numerical example illustrates, that it is often more informative to measure the sensitivity of demand to price on a *relative* rather than an *absolute* basis. Let us therefore generalize our numerical analysis. If a price of p changes by Δp then, as we saw in Exercise 11, the *relative* price change is given by

$$\frac{\Delta p}{p}.$$

And of course, the corresponding relative change in demand is

$$\frac{\Delta q}{q}.$$

To find out how these *relative* changes are related, the obvious starting point is our approximation for the *absolute* changes as described by the Δ-formula:

$$\Delta q \approx d'(p)\,\Delta p.$$

By dividing both sides by q, we find that

$$\frac{\Delta q}{q} \approx d'(p) \cdot \frac{1}{q}\,\Delta p \Longrightarrow \frac{\Delta q}{q} \approx d'(p) \cdot \frac{p}{q}\,\frac{\Delta p}{p}.$$

Finally, by using $q = d(p)$, we obtain that

$$\frac{\Delta q}{q} \approx d'(p) \cdot \frac{p}{d(p)}\,\frac{\Delta p}{p}.$$

The (red) factor which premultiplies the relative price change,

$$El_d(p) = d'(p) \cdot \frac{p}{d(p)},$$

elasticity

is called the *elasticity* of the demand function d. Using this notation, we get

$$\frac{\Delta q}{q} \approx El_d(p)\,\frac{\Delta p}{p}$$

Apparently,

the elasticity translates a small relative change in the price level into an approximation for the resulting relative change in demand.

The elasticity of the demand function should give us a reliable picture of the sensitivity of demand to price in the neighborhood of some price p. Moreover, the elasticity can quickly be calculated for all values of p, using the derivative function. Do bear in mind though that the elasticity only gives us an *approximation* for the relative change in the value of a function, just like the derivative gives you an approximation for the absolute change in the value of a function (remember Section 3).

In economics, the elasticity concept is of fundamental importance: not just to characterize the sensitivity of demand to price, but also e.g. the sensitivity of supply to price, of demand to consumer income, of demand w.r.t. the price of substitute products, etcetera, etcetera. Let us therefore define the elasticity concept for an *arbitrary* (positive) function.

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ELASTICITY

If $y = f(x)$, where f is a positive function, then the *elasticity* at some value $x > 0$ is defined as

$$\boldsymbol{El_f(x) = f'(x) \cdot \frac{x}{f(x)}}$$

The relative change $\dfrac{\Delta y}{y}$ in the variable y is approximately equal to $El_f(x)$ times the relative change $\dfrac{\Delta x}{x}$ in the variable x:

$$\boldsymbol{\frac{\Delta y}{y} \approx El_f(x)\,\frac{\Delta x}{x}}$$

We can use the definition above to obtain a general expression for the elasticity of our ferry-man's demand function, $d(p) = 1000(1 - \frac{1}{2}p + \frac{1}{16}p^2)$. Applying the rules from Section 2 to find the derivative of d, and plugging all the components correctly into the elasticity definition, we end up with

$$\begin{aligned} El_d(p) = d'(p) \cdot \frac{p}{d(p)} &= 1000(-\tfrac{1}{2} + \tfrac{1}{8}p) \cdot \frac{p}{1000(1 - \frac{1}{2}p + \frac{1}{16}p^2)} \\ &= \frac{-\frac{1}{2}p + \frac{1}{8}p^2}{1 - \frac{1}{2}p + \frac{1}{16}p^2} = \frac{2p^2 - 8p}{p^2 - 8p + 16}. \end{aligned}$$

You can easily verify that at a price of 2.50 euro, the formula above implies an elasticity of $El_d(2.5) = -3\frac{1}{3}$. In words: starting from an original price of 2.50 euro, a price increase by 1% causes an approximate decrease in the number of crossings by $3\frac{1}{3}$%. Consequently, a price increase by 10%, from 2.50 to 2.75 euro, causes a percentage change in crossings by roughly $El_d(2.5) \cdot 10 = -33.3\%$. This 33.3% decrease is a reasonable approximation for the exact decrease of 30.6% which we found before.

To round off this chapter, we will briefly apply the concept of 'the price elasticity of demand' to another example.

EXAMPLE 11 A DEMAND FUNCTION WITH CONSTANT ELASTICITY

In Example 4 of Chapter 2, the relation between demand D and price P was given by $D = d(P)$, where

$$d(P) = A\,P^{-\alpha}.$$

Note that d is a power function, with positive constants A and α. Since the derivative of this demand function is

$$d'(P) = -\alpha A\,P^{-\alpha-1},$$

its elasticity equals

$$El_d(P) = d'(P) \cdot \frac{P}{d(P)} = -\alpha A\,P^{-\alpha-1} \cdot \frac{P}{A\,P^{-\alpha}} = -\alpha.$$

Note: the price elasticity of this demand function is *constant*. This is a nontrivial result: e.g. in our ferry-man example, the price elasticity was visibly not constant (to convince yourself, you may try to determine the elasticity for $p = 3$ rather than $p = 2.5$). ◀

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END-OF-CHAPTER EXERCISES

■ **EXERCISE 1** Find the derivative of the function f defined by

(a) $f(x) = 5$ (b) $f(x) = x^4$ (c) $f(x) = 9x^{10}$ (d) $f(x) = \pi^7$

(e) $x^{-1}(x^2+1)\sqrt{x}$ (f) $\dfrac{1}{\sqrt{x^3}}$ (g) $\dfrac{x+1}{x^5}$ (h) $3x^{-11}$.

■ **EXERCISE 2** Find the (relative) rate of change and the elasticity of the function f defined by

(a) $f(r) = 4\pi r^2$ (b) $f(y) = Ay^{b+1}$ (c) $f(A) = \frac{1}{A^2\sqrt{A}}$.

■ **EXERCISE 3** Compute the elasticity of the power function f defined by $f(x) = x^a$.

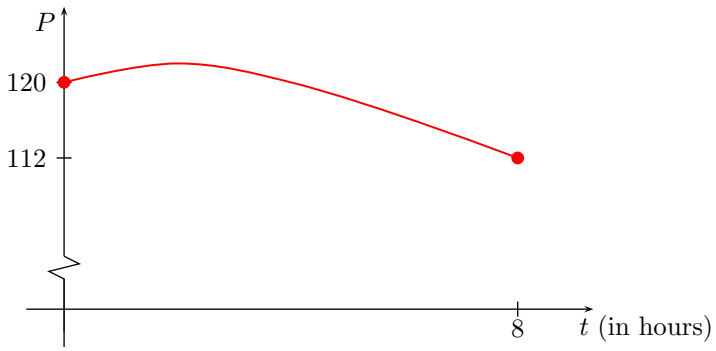
■ **EXERCISE 4** Prove that $El_{2f}(x) = El_f(x)$.

■ **EXERCISE 5** Find an equation of the tangent at the point $(0, 1)$ to the graph of the function f defined by

(a) $f(x) = (1+x)^{-1}$ (b) $f(x) = (1+x)^5$ (c) $f(x) = (1-x)^{1/4}$.

PRODUCTIVITY

An assembly plant has an 8-hour working day. An analysis of the production data has shown that, on a typical day, the productivity P in units per hour can be represented by the following graph:



Apparently, at the start of the working day, output is being produced at a rate of 120 units per hour, i.e. 2 units per minute. As the working day progresses, productivity is not constant: it fluctuates. So productivity can be regarded as a function of time t, say $P = p(t)$.

■ **EXERCISE 6** The total production during a working day can reasonably be estimated by

$$8 \times \frac{p(0) + p(8)}{2}$$

units. This estimate will be ...

(a) smaller than the real total production

(b) larger than the real total production

(c) exactly equal to the real total production

(d) There is not enough information to choose between the options (a), (b) and (c).

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The fluctuations in productivity during the working day, as displayed by our graph, can be described by the formula

$$p(t) = -\tfrac{1}{4}t^2 + t + 120.$$

■ **EXERCISE 7** The average rate of change of productivity during an entire working day is given by

(a) $\dfrac{p(0)+p(8)}{2}$

(b) $\dfrac{p'(0)+p'(8)}{2}$

(c) $\dfrac{p'(0)-p'(8)}{8}$.

(d) None of the options at (a), (b) or (c) is correct.

■ **EXERCISE 8** At the end of a working day, productivity is ...

(a) decreasing at a rate of $p'(8)$ units per minute

(b) decreasing at a rate of $p(8)$ units per hour

(c) decreasing at a rate of $p'(8)$ units per hour

(d) decreasing at a rate of $-p'(8)$ units per hour.

Management wants to know, how many units of output have been produced after t hours into the working day, for any t between 0 and 8. It can be shown that this number is given by the following expression:

$$Q = at^3 + \tfrac{1}{2}t^2 + 120t + b \qquad (0 \le t \le 8).$$

■ **EXERCISE 9** In this expression for Q, it must hold that ...

(a) $a < 0$ and $b > 0$

(b) $a = -\frac{1}{12}$ and $b < 0$

(c) $a = -\frac{1}{12}$ and $b = 0$

(d) $a = -\frac{3}{4}$ and $b = 0$.

GASOLINE

A gas station faces continuous price changes for its best-selling fuel type: Euro 95. The owner of the station has found out that the relation between the number of liters Q sold per day, and the price p in euros per liter, is given by

$$Q = 1000p^2 - 6000p + 12\,750.$$

■ **EXERCISE 10** If the price p (in euros per liter) increases by one cent, the resulting decrease in daily sales can be approximated by $b - ap$ liters. In this expression,

(a) $a < 15$ and $b < 50$

(b) $a > 15$ and $b < 50$

(c) $a < 15$ and $b > 50$

(d) $a > 15$ and $b > 50$.

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■ **EXERCISE 11** Currently, the price of a liter of Euro 95 is 150 cent, i.e. $p = 1.5$. How big is the price elasticity of demand?

(a) Somewhere between -0.2 and -0.4
(b) Somewhere between -0.4 and -0.6
(c) Somewhere between -0.6 and -0.8
(d) Somewhere between -0.8 and -1.0.

There has been a massive eruption of violence in the Middle East. The station owner expects that, as a result, the price of Euro 95 will shoot up from 150 to 175 cent next week.

■ **EXERCISE 12** Judged on the basis of the price elasticity of demand, what will happen to his revenue next week?

(a) Compared to this week, his revenue will go up
(b) Compared to this week, his revenue will go down
(c) His revenue will remain the same as this week
(d) There is not enough information to choose between the options (a), (b) and (c).

The station owner still expects the price of Euro 95 to rise from 150 to 175 cent per liter. He uses the concept of marginality to approximate the implied sales decrease next week. He then wonders how many customers he will lose per day, knowing that his average customer refuels around 50 liters of Euro 95 at a time.

■ **EXERCISE 13** According to the station owner's calculation, the number of customers will decrease by around ...

(a) 16 per day
(b) 15 per day
(c) 14 per day
(d) 13 per day.

AT WORK IN A FACTORY

The daily output Q of some factory is given by

$$Q = 900\, L^{1/3} \qquad (800 \leq L \leq 1200),$$

where L denotes the input of labour hours. Currently, 1000 labour hours are used each day.

A crucial feature of the function above is the so-called marginal product of labour; while we will not define this concept formally here, its meaning should be obvious from our discussion in the main text.

■ **EXERCISE 14** Currently, the marginal product of labour is around ...

(a) 2
(b) 3
(c) 4
(d) 5.

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EXERCISE 15 Consider the following two claims concerning the elasticity of the factory's output with respect to its labour input:

(I) Currently, this elasticity is around one-third.

(II) This elasticity has the same value for any level of the labour input.

(a) Claim (I) is correct, claim (II) is incorrect.

(b) Claim (I) is incorrect, claim (II) is correct.

(c) Both claims (I) and (II) are incorrect.

(d) Both claims (I) and (II) are correct.

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4 FINDING THE DERIVATIVE

In Section 1 of the previous chapter, we used a geometrical argument to explain the *meaning* of the derivative of a function f at some point a in its domain: it is the rate of change at that point, and it corresponds with the slope of the tangent to the curve of the function at that point. Initially, we used the *difference quotient* to calculate that derivative. This operation can obviously be performed for several (although maybe not all) points in the function's domain. In fact, we found that the derivative of a function, say f, was a new function: the *derivative function* f'. Of course, you would prefer to determine that derivative function (also called: to *differentiate* the original function) by using rules with general applicability, rather than to repeat the laborious 'difference quotient' approach. In Section 2, we reviewed three such rules which should be familiar from secondary school. First, we addressed an important *standard derivative*, allowing us to differentiate the class of power functions quickly, without re-inventing the wheel each time. Next, we covered two important *arithmetic rules*: the Sum Rule which allows us to take the derivative of a *sum* of functions, and the Simple Product Rule which enables us to differentiate a *multiple* of a function.

This week, we will discuss some more complicated arithmetic rules, and some additional standard derivatives.

We start with three crucial *arithmetic rules*: the (general) Product Rule and the Quotient Rule in Section 1, and the infamous Chain Rule in Section 2. Of course, many of you have already learned how to apply these rules in secondary school. But possibly your skills have become a bit rusty, while some of you may not have learned those rules at all. That's why we will carefully review them. At the end of Section 2, we will provide a complete list of all the arithmetic rules which are relevant for us.

In Section 3, we will discuss the *standard derivative* of two important classes of functions, introduced in Chapter 2: exponential and logarithmic functions. Again, for many of you this will boil down to a secondary school rehearsal. At the end of the section, we will provide a complete list of all the standard derivatives which are relevant for us.

1 THE PRODUCT RULE AND THE QUOTIENT RULE

Functions that can be regarded as the product or the quotient of basic functions can be differentiated by applying the Product Rule and the Quotient Rule. As already mentioned, you will probably recognize these rules from secondary school. If not, don't worry: we will provide plenty of examples. We start with the Product Rule.

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THE PRODUCT RULE

If the derivatives f' and g' exist, then

$$\big[f(x)\cdot g(x)\big]' = f'(x)\cdot g(x) + f(x)\cdot g'(x)$$

For those students who are not familiar with the derivative of a function, this rule may come somewhat 'out of the blue'; in particular, if you never saw the Product Rule before, it may strike you that there is no term involving $f'(x)\cdot g'(x)$. To motivate the structure of the rule, let's use the Δ-formula: if a is a number in the domain of a function h, then for a small change Δx (with respect to a) we have $\Delta y \approx h'(a)\,\Delta x$. Since $\Delta y = h(a+\Delta x) - h(a)$, we get

$$\begin{aligned}\Delta y \approx h'(a)\,\Delta x &\Longrightarrow h(a+\Delta x) - h(a) \approx h'(a)\ ,\Delta x\\ &\Longrightarrow h(a+\Delta x) \approx h(a) + h'(a)\,\Delta x\end{aligned}$$

Now consider a function which can be regarded as the product of two basic functions, say, $f(x)\cdot g(x)$. Applying our result above to this product, starting from some number a in the domain of both the functions f and g, we obtain:

$$\begin{aligned}f(a+\Delta x)\,g(a+\Delta x) &\approx \big[f(a)+f'(a)\,\Delta x\big]\big[g(a)+g'(a)\,\Delta x\big]\\ &= f(a)\,g(a) + f'(a)\,g(a)\,\Delta x + f(a)\,g'(a)\,\Delta x + f'(a)\,g'(a)\,\big(\Delta x\big)^2.\end{aligned}$$

If we now bring the term $f(a)\,g(a)$ to the left-hand side of the $\approx$ sign and divide both sides by Δx, we end up with

$$\frac{f(a+\Delta x)\,g(a+\Delta x) - f(a)\,g(a)}{\Delta x} \approx f'(a)\,g(a) + f(a)\,g'(a) + f'(a)\,g'(a)\,\Delta x.$$

Note: the left-hand side of this expression is the relevant difference quotient for finding the derivative of the function $f(x)\cdot g(x)$ at $x=a$! By choosing Δx closer and closer to zero, the last term in the right-hand side vanishes, and we are left with two terms which correspond with the formula in the Product Rule.

EXAMPLE 1 Consider the following function f, which is the product of two functions:

$$f(x) = (2x+1)(4x-3).$$

We can easily find the derivative of f using the Product Rule:

$$f'(x) = 2\cdot(4x-3) + (2x+1)\cdot 4 = 16x - 2.$$

Please note that we could also start by eliminating the brackets in the expression $(2x+1)(4x-3)$, to obtain $f(x) = 8x^2 - 2x - 3$; the derivative then follows (without the Product Rule!) again as $f'(x) = 16x - 2$.
So in this example, the Product Rule does not offer much of an advantage. ◀

We now want to show an interesting application of the Product Rule. In Chapter 3's Section 4, we introduced the *elasticity* concept. In Example 9 and its sequel, we used this concept to

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investigate the consequences of a price increase by a ferry-man. By calculating the elasticity for the ferry-man's demand function, we obtained an estimate for the percentage decrease in *demand* caused by a percentage price increase. But what about the consequences for his *revenue*? We already speculated a bit about this issue last week: now is the right time to formalize things.

EXAMPLE 2 THE PRICE ELASTICITY OF REVENUE

As we have seen, the price elasticity of *demand* measures how sensitive demand is to price changes: in particular, it measures the percentage change in demand in response to a percentage change in the price. By analogy, the price elasticity of *revenue* measures the percentage change in revenue in response to a percentage change in the price. Let's see what we can say about the price elasticity of revenue.

Of course, revenue R equals price times demand, which is itself a function of price, say $d(p)$; revenue is therefore also a function of price, say $R = r(p)$, with $r(p) = p \cdot d(p)$. Using the elasticity definition from Chapter 3, the price elasticity of revenue is given by

$$El_r(p) = r'(p) \cdot \frac{p}{r(p)}.$$

Of course, this expression contains the derivative of revenue with respect to the price, $r'(p)$. Since $r(p) = p \cdot d(p)$, we can use the Product Rule to find that $r'(p) = 1 \cdot d(p) + p \cdot d'(p)$. Substituting this into the elasticity definition and rearranging a bit, we get:

$$El_r(p) = [1 \cdot d(p) + p \cdot d'(p)]\frac{p}{p \cdot d(p)} = 1 + d'(p) \cdot \frac{p}{d(p)} = 1 + El_d(p).$$

We conclude: the price elasticity of revenue equals the price elasticity of demand, increased by 1. This is a pretty interesting result. Consider e.g. our ferry-man again. In last week's Example 9 and its sequel, we saw that at a price of 2.50 euro, the price elasticity of his *demand* equals $El_d(2.5) = -3\frac{1}{3}$. We now conclude that, at that price, the price elasticity of his *revenue* equals $El_r(2.5) = 1 + El_d(2.5) = -2\frac{1}{3}$. Consequently, a price increase by 10% will cause a revenue decrease of around $23\frac{1}{3}\%$. This result formalizes our earlier 'common sense' conclusion: raising his fare is not a good idea here. ◀

EXAMPLE 3 For a certain good, the relationship between the price p (currently 3 euro) and the demand in liters is aptly described by the demand function d defined by

$$d(p) = 2 + \frac{12}{p} \qquad (p > 0).$$

For this demand function, the price elasticity follows as

$$El_d(p) = d'(p) \cdot \frac{p}{d(p)} = \frac{-12}{p^2} \cdot \frac{p}{2 + 12/p} = -\frac{12}{2p + 12}.$$

So if the current price is $p = 3$, we have $El_d(3) = -\frac{2}{3}$. Now how will a price increase of e.g. 0.15 euro affect demand and revenue? This price increase amounts to 5%, so the percentage change in *demand* can be approximated by $El_d(3) \cdot 5 = -\frac{2}{3} \cdot 5 = -3\frac{1}{3}\%$. Moreover, with $El_r(3) = 1 + El_d(3) = \frac{1}{3}$, the percentage change in *revenue* should be approximately $\frac{1}{3} \cdot 5 = 1\frac{2}{3}\%$. Note that, since the price elasticity of revenue is positive now, revenue rises when the price is increased.

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Please verify that the *exact* revenue increase (from 18 to 18.3) equals 1.666% here. The fact that our approximation coincides with the exact value is a matter of pure coincidence. ◀

Note: In our ferry-man example, the price elasticity of demand was *smaller* (in the sense of 'more negative') than -1; consequently, the price elasticity of revenue was *negative*. So if the price is raised, the relative decline in demand will be *larger* than the relative price rise, and the net result is that revenue will *decrease*. In Example 3 above, the price elasticity of demand was *larger* (in the sense of 'less negative') than -1; consequently, the price elasticity of revenue was *positive*. So in case of a price increase, the relative decline in demand will be *smaller* than the relative price rise, and the net result is that revenue will *increase*.

An interesting borderline scenario occurs when, at some price level, the price elasticity of demand is exactly *equal* to -1. In this so-called 'unit elasticity' case, the price elasticity of revenue equals 0. This implies that small price changes, triggering proportional demand changes, will leave revenue approximately *unchanged*; so apparently, revenue reaches its *maximum* at that price.

Having addressed the Product Rule, and its application to the price elasticity of revenue, we now move to the Quotient Rule.

THE QUOTIENT RULE

If the derivatives f' and g' exist, then

$$\left[\frac{f(x)}{g(x)}\right]' = \frac{g(x)\cdot f'(x) - f(x)\cdot g'(x)}{g(x)^2}$$

Again, in the absence of vivid memories from secondary school, this rule will come 'out of the blue', so we want to motivate its structure. As it turns out, the Quotient Rule can actually be derived from the preceding Product Rule. Our interest is now in the derivative of a function which can be regarded as the *quotient* (rather than the product) of two functions, say $h(x) = f(x)/g(x)$. When we multiply both sides by $g(x)$, this translates into

$$f(x) = h(x)\cdot g(x).$$

If we now apply the Product Rule, we obtain:

$$f'(x) = h'(x)\cdot g(x) + h(x)\cdot g'(x).$$

This equation can be solved for the sought-after derivative of h:

$$h'(x) = \frac{f'(x)}{g(x)} - \frac{h(x)\cdot g'(x)}{g(x)}.$$

Finally, if we replace $h(x)$ in the right-hand side by its definition $f(x)/g(x)$ then, after some rewriting, we indeed end up with the Quotient Rule:

$$h'(x) = \frac{f'(x)}{g(x)} - \frac{f(x)\cdot g'(x)}{g(x)^2} = \frac{g(x)\cdot f'(x) - f(x)\cdot g'(x)}{g(x)^2}.$$

In the following example, as in the previous Example 1, there are again (at least) two ways of dealing with the problem.

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EXAMPLE 4 Let

$$h(x) = \frac{3x^2 - 5x}{\sqrt{x}} \qquad (x > 0).$$

We can view this function as the quotient of $f(x) = 3x^2 - 5x$ and $g(x) = \sqrt{x} = x^{1/2}$. Applying the Quotient Rule leads to:

$$\begin{aligned} h'(x) &= \frac{x^{1/2} \cdot (6x - 5) - (3x^2 - 5x) \cdot \frac{1}{2}x^{-1/2}}{x} \\ &= 6x^{1/2} - 5x^{-1/2} - \tfrac{3}{2}x^{1/2} + \tfrac{5}{2}x^{-1/2} = \tfrac{9}{2}\sqrt{x} - \frac{5}{2\sqrt{x}}. \end{aligned}$$

Alternatively, we can rewrite our function as $h(x) = (3x^2 - 5x) \cdot x^{-1/2} = 3x^{3/2} - 5x^{1/2}$, so that it can be differentiated without using the Quotient Rule:

$$h'(x) = \tfrac{9}{2}x^{1/2} - \tfrac{5}{2}x^{-1/2} = \tfrac{9}{2}\sqrt{x} - \frac{5}{2\sqrt{x}}.$$

As you can see, both approaches again give the same result (of course!). ◀

■ **EXERCISE 1** Determine the derivative of the function h in two different ways: *first* by using the Product or Quotient Rule, and *next* by eliminating the brackets or developing the ratio. Which method is easier and/or faster here?

(a) $h(x) = \left(\sqrt{x} + x\sqrt{x}\right)\dfrac{1}{x^2\sqrt{x}}$ (b) $h(x) = (1 - x)\left(x^3 + \dfrac{1}{x^3}\right)$.

■ **EXERCISE 2** Remember Chapter 2's Example 2, where the total cost of the Toyota dealer could be expressed as a broken function k of order size q, with

$$k(q) = \frac{18\,000 + 500q^2}{q} \qquad (q > 0).$$

Find the derivative of the function k in two different ways: *first* by using the Quotient Rule, and *next* by developing the ratio. Which method is easier and/or faster here?

From Examples 1 and 4 and Exercises 1 and 2 we can conclude that, even when a function can be viewed as a product or a quotient of functions, it may not be necessary (or desirable) to use the Product Rule or the Quotient Rule. But in other cases, those rules are indispensable. The next example shows this for the Quotient Rule.

EXAMPLE 5 Consider the function h defined by

$$h(x) = \frac{x + 1}{x^2 + 2x} \qquad (x \neq 0 \text{ and } x \neq -2).$$

This time, we *have* to use the Quotient Rule: there is nothing to gain from 'developing the ratio' here. The function h can be viewed as the quotient of the two functions f and g, defined by $f(x) = x + 1$ and $g(x) = x^2 + 2x$. Since $f'(x) = 1$ and $g'(x) = 2x + 2$, the derivative of h follows as:

$$h'(x) = \frac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{g(x)^2} = \frac{(x^2 + 2x) \cdot 1 - (x + 1) \cdot (2x + 2)}{(x^2 + 2x)^2} = \frac{-x^2 - 2x - 2}{(x^2 + 2x)^2}. \quad ◀$$

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■ **EXERCISE 3** Find the derivative of the function f defined by $f(x) =$

$$\text{(a)}\ \left(\frac{1}{x}+1\right)(x^2+1) \qquad \text{(b)}\ \frac{x+1}{x-1} \qquad \text{(c)}\ \frac{\sqrt{x}+1}{\sqrt{x}-1}.$$

2 THE CHAIN RULE

In Chapter 2's Example 5, we studied the construction of a triangular gutter with maximal capacity out of a plate of length 100 and width 18. We found that, if x (instead of the original symbol s) represents the length of the base of the triangle, then the capacity of the gutter is given by $C = c(x)$, where

$$c(x) = 50x\sqrt{81 - \tfrac{1}{4}x^2} \qquad (0 < x \leq 18).$$

Your secondary school memory (which will be recapitulated in the next chapter) should tell you that, in order to find the value of x which *maximizes* the capacity of the gutter, we will need the derivative of the function c. Of course, c can be regarded as the product of two functions of x: this suggests the use of the Product Rule. However, there is an obvious complication here. In the last factor of the function c, under the square root sign, we do not simply have x, but a (quadratic) *function* of x. As will be defined formally below, such a structure is called a *composite function*. To differentiate such a function, we need the *Chain Rule*.

To bring out the issues as clearly as possible, let us focus on the 'tricky' part of our capacity function, i.e. its last factor, to which we will refer as function k:

$$k(x) = \sqrt{81 - \tfrac{1}{4}x^2}.$$

Now have a very careful look at the structure of this function. Imagine that you want to calculate $k(x)$ for a given value of x, using your pocket calculator. You will definitely do this in *two* successive steps: *first* you calculate $81 - \frac{1}{4}x^2$, and *then* you take the square root of your previous result. Generalizing, the value $k(x)$ is obtained by *first* calculating the value $g(x) = 81 - \frac{1}{4}x^2$, and *then* applying to this result the square root function f defined by $f(y) = \sqrt{y}$. Consequently, the function k can actually be written as $k(x) = f\big((g(x)\big)$. A function which can be written in this way is called a *composite function*.

COMPOSITE FUNCTIONS

A function h with the structure $h(x) = f(g(x))$ is called a *composite function*. Since g is applied first, and f is then applied to the result, we also speak of the function 'f after g'. The function g is called the *inner* function, while f is called the *outer* function.

EXAMPLE 6 The expression $h(x) = (2x+3)^4$ can be written as $f(g(x))$, with $g(x) = 2x+3$ and $f(y) = y^4$. The function h is therefore composite.
The expression $m(x) = \sqrt{x^3}$ can be written as $f(g(x))$, with $g(x) = x^3$ and $f(y) = \sqrt{y}$. The function m is therefore also composite. ◀

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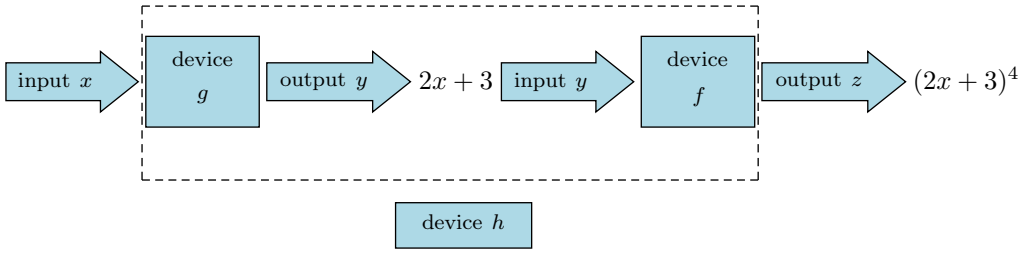
The input-output interpretation of a function which we presented in Chapter 1 can be helpful to understand the idea of a composite function. Consider the first composite function discussed in Example 6: $h(x) = (2x+3)^4$. We have seen that we can write this function as $h(x) = f(g(x))$, with 'inner' function $g(x) = 2x+3$, and 'outer' function $f(y) = y^4$. In the remainder, we will refer to the 'output' of the function h as z, and to the 'output' of the function g as y.



The function h can be interpreted as a device, or machine, which converts an input value x into an output value of $z = (2x+3)^4$. This conversion can actually be understood as the result of two successive operations:

– first, machine g converts the input x into an *intermediate* output of $y = 2x+3$;

– then machine f converts $y = 2x+3$ into the final output of $z = y^4 = (2x+3)^4$.

In the figure above, we have put in line the two machines that correspond with the functions f and g. The crucial observation is that the output of the first machine, $2x+3$, serves as the input for the second one. Note that the order of the two machines is relevant here! Imagine that we interchange the two machines in our 'assembly line'; i.e. f is applied first, and then g is applied to the result. In that case, we get a very different expression (check!):

$$x \to x^4 \to 2x^4 + 3.$$

Now how to find the derivative of such a composite function? As already stated, this requires the dreaded Chain Rule. We will motivate this rule in a similar way as for the Product Rule in the previous section. Conceptually, we take the picture above as our starting point. So: consider some value for x, say a, in the domain of an (unspecified) composite function $h(x) = f(g(x))$. Let's examine the effect of a small change in x on the value of h, i.e. on z. We can think of this effect in two alternative (but ultimately equivalent) ways.

the direct way

First, let us *not* 'open up' machine h. Of course, the derivative of h translates the change in x into an approximate change in z. This is the Δ-formula from Chapter 3's Section 3, which we can formally express as follows:

$$\Delta z \approx h'(a)\,\Delta x. \tag{1}$$

in two steps

Next, we *do* 'open up' machine h. We can now determine the change in z in two steps. *First*, the presence of machine g implies that, starting from $x = a$, a small change in x will trigger a change in y. And of course, the derivative of g determines how large that change will be, approximately. Formally:

$$\Delta y \approx g'(a)\,\Delta x.$$

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Second, the presence of machine f implies that, starting from $y = b$, a small change in y will create a change in z. The approximate size of the change depends on the derivative of f:

$$\Delta z \approx f'(b)\,\Delta y.$$

Since the input of machine f is the output of machine g, we have $b = g(a)$. Successive substitution leads to

$$\Delta z \approx f'(b) \cdot \Delta y \approx f'\big(g(a)\big) \cdot \Delta y \approx f'\big(g(a)\big) \cdot g'(a) \cdot \Delta x. \tag{2}$$

If you now compare equations (1) and (2), then the formal rule stated below should not come as a surprise.

THE CHAIN RULE

If the derivatives f' and g' exist, then

$$\big[f\big(g(x)\big)\big]' = f'\big(g(x)\big) \cdot g'(x)$$

So although the Chain Rule may initially look very abstract, it actually makes perfect intuitive sense:

– a change in x by one unit triggers a change in y by roughly g' units;

– every one unit change in y then triggers a change in z by roughly f' units;

– the ultimate change in z is therefore the product of those two derivatives.

Now how to apply the chain rule in practice? As you may remember from secondary school, an obvious approach is to apply the rule in two steps:

step 1: determine $f'(y)$ and $g'(x)$.

step 2: in the expression $f'(y) \cdot g'(x)$, replace y by $g(x)$.

EXAMPLE 7 Consider again the composite functions introduced in Example 6. As we have seen, the expression $h(x) = (2x+3)^4$ can be written as $f(g(x))$, with $g(x) = 2x+3$ and $f(y) = y^4$. So

step 1: $f'(y) = 4y^3$ and $g'(x) = 2$

step 2: according to the Chain Rule we have

$$h'(x) = f'(g(x)) \cdot g'(x) = 4(g(x))^3 \cdot 2 = 4(2x+3)^3 \cdot 2 = 8(2x+3)^3.$$

The expression $m(x) = \sqrt{x^3}$ can be written as $f(g(x))$, with $g(x) = x^3$ and $f(y) = \sqrt{y}$. So

step 1: $f'(y) = \frac{1}{2\sqrt{y}}$ and $g'(x) = 3x^2$

step 2: according to the Chain Rule we have

$$m'(x) = f'(g(x)) \cdot g'(x) = \frac{1}{2\sqrt{g(x)}} \cdot 3x^2 = \frac{1}{2\sqrt{x^3}} \cdot 3x^2.$$

◀

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You may find the *second* part of Example 7 a bit unconvincing; after all, there is an obvious temptation to rewrite the function m as $m(x) = x^{3/2}$, and then we don't need the Chain Rule anymore to obtain its derivative! But in many cases the Chain Rule is truly *indispensable*, as demonstrated in Example 8 below, and in Exercises 4 and 5.

EXAMPLE 8 A CONSTRUCTION PROBLEM

Let's return to the problem which actually triggered our Chain Rule discussion: the construction of a triangular gutter with maximal capacity. Remember: with x representing the length of the base of the triangle, the gutter's capacity is given by $C = c(x)$, where

$$c(x) = 50x\sqrt{81 - \tfrac{1}{4}x^2} \qquad (0 < x \leq 18).$$

We have already established that, to differentiate this capacity function, we will need to use both the Product Rule **and** the Chain Rule. The result is:

$$c'(x) = 50\sqrt{81 - \tfrac{1}{4}x^2} + 50x \cdot \frac{1}{2\sqrt{81 - \frac{1}{4}x^2}} \cdot -\tfrac{1}{2}x = 50\sqrt{81 - \tfrac{1}{4}x^2} - \frac{25x^2}{2\sqrt{81 - \frac{1}{4}x^2}}. \quad \blacktriangleleft$$

■ **EXERCISE 4** Determine the derivative of the function h defined by

(a) $h(x) = \sqrt{2x^2 + 1}$ (b) $h(x) = (3x-1)^6$ (c) $h(x) = \dfrac{1 - \sqrt{2x}}{1 + \sqrt{2x}}$ $(x > 0)$.

■ **EXERCISE 5** Let f be a function for which the derivative exists. Find the derivative of

(a) $f(x^2) + x^2$ (b) $xf(x)$ (c) $\big[f(x)\big]^2$ (d) $\dfrac{x}{f(x)}$ (e) $\sqrt{f(x)}$.

To conclude this section, we reiterate all the *arithmetic rules* which have been discussed here and in the previous chapter.

LIST OF ARITHMETIC RULES

name of the rule	
sum rule	$\big[f(x) + g(x)\big]' = f'(x) + g'(x)$
simple product rule	$\big[cf(x)\big]' = cf'(x)$
product rule	$\big[f(x) \cdot g(x)\big]' = f'(x) \cdot g(x) + f(x) \cdot g'(x)$
quotient rule	$\left[\dfrac{f(x)}{g(x)}\right]' = \dfrac{g(x) \cdot f'(x) - f(x) \cdot g'(x)}{g(x)^2}$
chain rule	$\big[f\big(g(x)\big)\big]' = f'(g(x)) \cdot g'(x)$

3 THE DERIVATIVE OF EXPONENTIAL AND LOGARITHMIC FUNCTIONS

In Chapter 2 we have seen examples of growth (or decay) at a constant rate, like the growth of China's GDP per head, and the depreciation of the value of a car. In all these cases we

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were able to describe the process by means of *exponential* functions. Our discussion then naturally carried us to their inverse: *logarithmic* functions. In this section we will study the derivatives of both exponential and logarithmic functions.

3.1 THE DERIVATIVE OF EXPONENTIAL FUNCTIONS

Let's consider the exponential function f of the form $f(x) = a^x$, with $a > 0$. Since $f(0) = a^0 = 1$, $(0, 1)$ is a point on the graph of this function, for *any* base value a. Although our ultimate objective is to find the derivative of the entire function, we start by investigating its derivative at this specific point, i.e. for $x = 0$.

THE DERIVATIVE OF EXPONENTIAL FUNCTIONS AT $x = 0$

As an example, take the exponential function with base 2. What can we say about its derivative at $x = 0$? Well, if $g(x) = 2^x$, the relevant difference quotient is

$$\frac{g(0 + \Delta x) - g(0)}{\Delta x} = \frac{2^{\Delta x} - 2^0}{\Delta x} = \frac{2^{\Delta x} - 1}{\Delta x}.$$

In Table 1 below, we evaluate this difference quotient for a number of values for Δx which come closer and closer to zero.

Δx	$\dfrac{2^{\Delta x} - 1}{\Delta x}$
0.1	0.71773
0.01	0.69556
0.001	0.69339
0.0001	0.69317
0.00001	0.69315

TABLE 1 The behaviour of the difference quotient as Δx gets smaller

This table suggests that the limiting value of our difference quotient is

$$\lim_{\Delta x \to 0} \frac{2^{\Delta x} - 1}{\Delta x} \approx 0.69315,$$

implying that the derivative of g at $x = 0$ is given by

$$g'(0) \approx 0.69315 < 1.$$

■ **EXERCISE 6** Consider the exponential function h defined by $h(x) = 3^x$.
Make a table similar to Table 1 and show that $h'(0) \approx 1.09862 > 1$.

As you know, the number $g'(0)$ represents the slope of the tangent (say ℓ) to the curve $y = 2^x$ at the point $(0, 1)$. Similarly, the number $h'(0)$ represents the slope of the tangent (say ℓ') to the curve $y = 3^x$ at that same point $(0, 1)$. Both these tangents are sketched in Figure 1 below.

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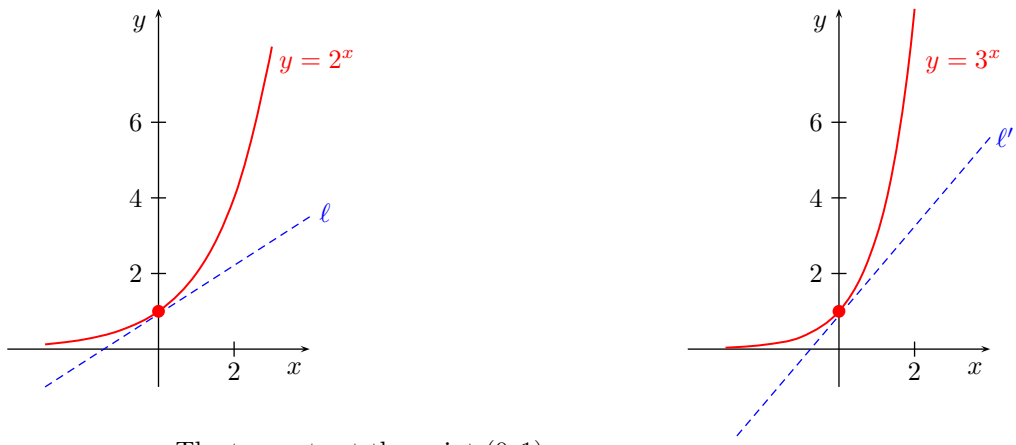

FIGURE 1 The tangents at the point $(0, 1)$

If you compare the two tangents ℓ and ℓ' in Figure 1, it is visible that the angle of tangent ℓ with the horizontal axis is *smaller* than 45 degrees, while tangent ℓ' has an angle with the horizontal axis which is *larger* than 45 degrees. This corresponds with the fact that the slope of the first tangent (around 0.69315) is *smaller* than 1, while the slope of the second tangent (roughly 1.09862) is *larger* than 1. Apparently, when the base of our exponential function increases (here: from 2 to 3), then the slope of the tangent to its graph at the point $(0, 1)$ increases. But then there must be a number somewhere between 2 and 3, such that the corresponding exponential function has a tangent with slope *equal* to 1. This number will be denoted by e (named after the famous mathematician Leonhard Euler). The number e, which is approximately equal to 2.718, plays an important role in a lot of economic applications.

Summarizing: the number e is defined such, that the slope of the tangent to the curve $y = \mathrm{e}^x$ at the point $(0, 1)$ equals 1. Formally:

If $f(x) = \mathrm{e}^x$, then

$$f'(0) = \lim_{\Delta x \to 0} \frac{\mathrm{e}^{\Delta x} - 1}{\Delta x} = 1.$$

THE DERIVATIVE OF THE EXPONENTIAL FUNCTION WITH BASE e

The result which we have just obtained may seem quite useless: after all, it concerns the derivative of one very specific exponential function (with base e), at one very specific value of x (i.e. $x = 0$). But on the basis of this result, it is actually possible to find the derivative of *any* simple exponential function.

To show this, we start with the exponential function with base e: $f(x) = \mathrm{e}^x$. If we want to obtain the derivative of this function at some value of x (other than $x = 0$), then the relevant difference quotient is

$$\frac{f(x + \Delta x) - f(x)}{\Delta x} = \frac{\mathrm{e}^{x+\Delta x} - \mathrm{e}^x}{\Delta x} = \frac{\mathrm{e}^x \cdot \mathrm{e}^{\Delta x} - \mathrm{e}^x}{\Delta x} = \mathrm{e}^x \cdot \frac{\mathrm{e}^{\Delta x} - 1}{\Delta x}.$$

As always, we find the derivative of f by evaluating what happens to this difference quotient if Δx gets closer and closer to zero. Using the fact that $\dfrac{\mathrm{e}^{\Delta x} - 1}{\Delta x} \to 1$ as $\Delta x \to 0$, we can

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easily conclude that

$$f'(x) = \mathrm{e}^x.$$

We have now found the following important result:

THE DERIVATIVE OF THE EXPONENTIAL FUNCTION WITH BASE e

$$\left[\mathrm{e}^x\right]' = \mathrm{e}^x$$

By using the Arithmetic Rules discussed before, it is possible to differentiate more complex exponential functions with base e, as the following example illustrates.

EXAMPLE 9 Let the function h be defined by

$$h(x) = \mathrm{e}^{x^2+x}.$$

Since $h(x)$ can be written as $f\big(g(x)\big)$ with $g(x) = x^2 + x$ and $f(y) = \mathrm{e}^y$, we conclude that h is a composite function. We can therefore find its derivative by applying the Chain Rule:

$$h'(x) = \mathrm{e}^{x^2+x} \cdot (2x+1) = (2x+1)\,\mathrm{e}^{x^2+x}.$$ ◀

■ **EXERCISE 7** Find the derivative of the function h defined by

(a) $h(x) = x\,\mathrm{e}^{x^2}$ (b) $h(x) = \dfrac{x}{\mathrm{e}^x}$ (c) $h(x) = 5\,\mathrm{e}^{5\sqrt{x}}$.

THE DERIVATIVE OF GENERAL EXPONENTIAL FUNCTIONS

Now consider the derivative of the general exponential function f defined by

$$f(x) = a^x,$$

where the base doesn't have to be the number e: it can be any $a > 0$. As it turns out, the derivative of f can be obtained by cleverly rewriting the function. To do so, we need some of the properties of logarithms which were discussed in Chapter 2's Section 3.2. Remember that the solution of the *exponential equation*

$$b^x = c$$

(with $c > 0$) is given by ${}^b\log c$. We also learned how to express this logarithm with base b in terms of the quotient of two logarithms with base 10. This result was important, because the logarithm with base 10 can be found under the *log key* of your pocket calculator. But in many cases, it is actually preferable to use logarithms with the base number e (instead of 10). This so-called *natural logarithm* is denoted by ln (so $\ln x = {}^{\mathrm{e}}\log x$), and can be found on your pocket calculator under the *ln key*. For example, you can use that button to check that $\ln 2 \approx 0.69315$. So, by definition, the solution of the exponential equation

natural logarithm

$$\mathrm{e}^x = a$$

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is given by ${}^{\mathrm{e}}\log a = \ln a$. Make sure to remember what this means in words: $\ln a$ is 'the exponent to which we must raise e, in order to obtain a'. In other words:

$$\mathrm{e}^{\ln a} = a$$

Let's now return to our general exponential function f. If we replace its base a by the equivalent $\mathrm{e}^{\ln a}$, we get:

$$f(x) = a^x = \left[\mathrm{e}^{\ln a}\right]^x = \mathrm{e}^{x \ln a}.$$

Using this 'trick', we have managed to rewrite the general exponential function f as an exponential function with base e. Applying the Chain Rule, it's now trivial to find its derivative:

$$f'(x) = \mathrm{e}^{x \ln a} \cdot \ln a = a^x \cdot \ln a.$$

We have now found the following property:

THE DERIVATIVE OF EXPONENTIAL FUNCTIONS WITH ANY BASE

For $a > 0$,

$$\left[a^x\right]' = a^x \ln a$$

EXAMPLE 10 Let the function h be defined by

$$h(x) = 2^{x^2+x}.$$

Since $h(x)$ can be written as $f\big(g(x)\big)$ with $g(x) = x^2 + x$ and $f(y) = 2^y$, the function h is composite. We can therefore obtain its derivative by applying the Chain Rule:

$$h'(x) = 2^{x^2+x} \ln 2 \cdot (2x+1) = (2x+1)\, 2^{x^2+x} \ln 2.$$ ◀

EXAMPLE 11 EXPONENTIAL GROWTH

In Chapter 2's Section 3, we observed that exponential growth refers to processes involving, quote, 'quantities that increase or decrease by amounts which are proportional to their size'. If an exponential function, say $f(t) = b \cdot a^t$, is used to describe such an exponential growth process then, following Chapter 3, its derivative f' represents the *rate of change* of that process. Hence, we obtain:

$$f'(t) = b \cdot a^t \ln a = (\ln a) \cdot f(t).$$

It should be clear that this result is totally consistent with the defining characteristic of exponential growth as embodied in our quote from Chapter 2 above! To make things a bit more specific:

> *If a quantity grows exponentially, then the rate of change is at every moment proportional to the value of the quantity at that moment.* ◀

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■ **EXERCISE 8** On January 1st 2015, a certain area is home to a population of 100 badgers. On January 1st 2018, their number has increased to 108. A biologist assumes that the number of badgers in the area grows exponentially. Let's denote the number of badgers after t years from January 1st 2015 ($t = 0$) by $N(t)$.

(a) Determine the exponential function N.

(b) What is the growth factor of the badger population per 10 years?

(c) Determine the 'time to double', i.e. the amount of time it takes for the badger population to become twice as large.

(d) Determine the rate of change of the number of badgers on January 1st 2018.

■ **EXERCISE 9** A machine cost 10 000 euro at the time it was purchased ($t = 0$). Two years later, its value has dropped to only 7000 euro. We assume that the decrease in value evolves over time according to an exponential function. Let's denote the machine's value precisely t years after the time of purchase by $V(t)$.

(a) Determine the exponential function V.

(b) What is the growth factor per half year?

(c) Determine the 'halflife value', i.e. the amount of time it takes for the machine's value to be cut in half.

(d) Determine the rate of change of the machine's value, 5 years after its purchase.

3.2 THE DERIVATIVE OF LOGARITHMIC FUNCTIONS

As our last topic this week, we will show you how to determine the derivative of the logarithmic function with base $a > 0$. So: consider the function f defined by

$$f(x) = {}^{a}\log x \quad (x > 0).$$

By definition, it holds (for any positive x) that

$$a^{f(x)} = a^{{}^{a}\log x} = x.$$

If you do not see this immediately, then remember that ${}^{a}\log x$ is 'the exponent to which we must raise a, in order to obtain x'. Now let's see what happens when we differentiate both the left- and the right-hand side of this equation with respect to x. To find the derivative of the *left*-hand side, we need to apply the Chain Rule. The result is:

$$\left[a^{f(x)}\right]' = a^{f(x)} \ln a \cdot f'(x) = a^{{}^{a}\log x} \ln a \cdot f'(x) = x \ln a \cdot f'(x).$$

The derivative of the *right*-hand side, x, is of course 1. Since these two derivatives must be equal, we get:

$$x \ln a \cdot f'(x) = 1.$$

We can now solve this equation to obtain the derivative of f:

$$f'(x) = \frac{1}{x \ln a}.$$

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THE DERIVATIVE OF LOGARITHMIC FUNCTIONS

For $a > 0$ and $x > 0$,

$$\left[{}^{a}\log x\right]' = \frac{1}{x \ln a}$$

Finally, we consider an important special case of our result: choose $a = \mathrm{e}$. Using the fact that $\ln \mathrm{e} = 1$, we now find that, for $x > 0$,

$$\left[\ln x\right]' = \frac{1}{x}$$

■ **EXERCISE 10** Determine the derivative of the function f defined by

(a) $f(t) = 10^t$ (b) $f(t) = 5 \log t$ (c) $f(K) = \sqrt{K} \cdot 3^{0.1K}$

(d) $f(t) = \ln 6t$ (e) $f(t) = 1000(1 - \mathrm{e}^{-0.2t})$ (f) $f(t) = \dfrac{1000}{1 + 500\,\mathrm{e}^{-0.2t}}$

■ **EXERCISE 11** Find the derivative of the function f defined by $f(x) =$

(a) $x + \ln x$ (b) $\mathrm{e}^x \ln x$ (c) $\ln\left(\dfrac{x+1}{x-1}\right)$ (d) $\dfrac{x}{\ln x}$

To conclude this section, we reiterate all the *standard derivatives* which have been discussed here and in the previous chapter.

LIST OF STANDARD DERIVATIVES

$f(x) =$		$f'(x) =$
c	(c is some number)	0
x^r	(r is some real number)	rx^{r-1}
e^x		e^x
a^x	($a > 0$)	$a^x \ln a$
$\ln x$		$\dfrac{1}{x}$
${}^{a}\log x$	($a > 0$)	$\dfrac{1}{x \ln a}$

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END-OF-CHAPTER EXERCISES

EXERCISE 1 Differentiate the following expressions

(a) $\dfrac{x-1}{x+1}$ (b) $\dfrac{x-1}{x^2-1}$ (c) $\dfrac{3x-1}{x^2+x+1}$

(d) $\dfrac{ax+b}{cx+d}$ (e) $x^n\big(a\sqrt{x}+b\big)$ (f) $\dfrac{1}{ax^2+bx+c}$

EXERCISE 2 Find a formula for $h'(x)$ if

(a) $h(x) = f(x^2)$ (b) $h(x) = f\big(x^n g(x)\big)$.

EXERCISE 3 Find $El_x f(x)$ if

(a) $f(x) = \ln x$

(b) $f(x) = x^p\,\mathrm{e}^{ax}$ (here a and p are constants).

EXERCISE 4 The demand for D for apples in the U.S. as a function of income r for the period 1927 to 1941 was estimated as $D = A\,r^{1.23}$, where A is a constant. Find and interpret the elasticity of D with respect to r, ie. the income elasticity of demand.

EXERCISE 5 A wine farmer sells wine to customers who visit the farm. Suppose that the daily demand d in liters as function of the price p in euros per liter is given by

$$d(p) = \frac{40}{\sqrt{4p^2+4p+1}} \qquad (p \geq 0).$$

At a price of 6 euros per liter, the price is decreased with 10 cents. With help of the (point) price elasticity we can conclude that, as a consequence of the price decrease, the demand increases or decreases with how many percent?

AN INSTALLATION PROBLEM

In a number of end-of-chapter exercises of Chapter 2, we studied the installation costs of a water supply pipe running from a residence to a country cottage. We found that, if x (rather than ℓ) represents the length of the pipe alongside the asphalted road, these costs amount to $K = k(x)$, where

$$k(x) = 40x + 80\sqrt{x^2 - 400x + 80\,000} \qquad (0 \leq x \leq 200).$$

EXERCISE 6 Taking the derivative of this cost function leads to $k'(x) =$

(a) $40 + \dfrac{80(x-200)}{\sqrt{x^2-400x+80\,000}}$

(b) $40 + \dfrac{80(2x-400)}{\sqrt{x^2-400x+80\,000}}$

(c) $40 + \dfrac{40(x-200)}{\sqrt{x^2-400x+80\,000}}$.

(d) None of the options (a), (b) or (c) is correct.

Next week we will see that, when we want to find the minimum of the function k, the zero(s) of its derivative function k' are of great interest.

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■ **EXERCISE 7** The zero(s) of the function k' can be found by solving the quadratic equation

(a) $3x^2 + 1200x + 80\,000 = 0$

(b) $3x^2 - 1200x + 80\,000 = 0$

(c) $3x^2 + 1200x - 80\,000 = 0$

(d) $3x^2 - 1200x - 80\,000 = 0$.

A FENCE

In Example 5 of Chapter 1 and its sequel, Example 1 of Chapter 2, we studied the problem of fencing a piece of land, where the fenced area should be 100 m^2 large. We found that, if x (rather than ℓ) represents the length of the fenced area, then the total length L of the fence is given by $L = \ell(x)$, where

$$\ell(x) = \frac{2x^2 + 200}{x} \qquad (x > 0).$$

The challenge was to minimize the length of the fence.

■ **EXERCISE 8** Taking the derivative of this function leads to $\ell'(x) =$

(a) $\dfrac{200}{x^2} + 6$

(b) $2 - \dfrac{200}{x^2}$

(c) $2 + \dfrac{200}{x^2}$

(d) $4x$.

■ **EXERCISE 9** If a is the unique zero of the function ℓ', then $\ell(a) =$

(a) 10

(b) 20

(c) 30

(d) 40.

LOGISTIC GROWTH

The way in which the sales S of a certain product develop over time can be described by a *logistic growth* function. That is: $S = s(t)$, with

$$s(t) = c + \frac{1}{d + \mathrm{e}^{-2t}} \qquad (t \geq 0),$$

where $c > 0$ and $d \in (0, 1)$ are constants.

■ **EXERCISE 10** In the long run (that is: for large values of t), sales S will approach ...

(a) c

(b) $c + \dfrac{1}{d+1}$

(c) $\dfrac{1 + cd}{d}$

(d) $\dfrac{c}{d}$.

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■ **EXERCISE 11** The derivative $s'(t)$ is ...

(a) positive for all values of $t \geq 0$

(b) first positive and then negative

(c) first positive, then negative and then positive again

(d) negative for all values of $t \geq 0$.

Obviously, the derivative s' which you determined in the previous exercise is itself a function of time. Therefore, it is possible to calculate the derivative of s'.

■ **EXERCISE 12** The derivative of the function s' is zero for

(a) $t = \frac{1}{2} \ln d$

(b) $t = -\frac{1}{2} \ln d$

(c) $t = \frac{1}{2} d$

(d) $t = -\frac{1}{2} d$.

■ **EXERCISE 13** At $t = 0$, the sales level S is 6, and its rate of change is $1\frac{1}{8}$. What are the values of c and d?

(a) $c = 5$ and $d = 0$

(b) $c = 5.2$ and $d = 0.25$

(c) $c = 5.25$ and $d = \frac{1}{3}$

(d) $c = 5.5$ and $d = 1$.

THE DEMAND FOR GASOLINE

Suppose that the demand D (in liters) for gasoline at price p (in euros) is given by

$$D = b\,\mathrm{e}^{-0.2p},$$

where b is a constant. Clearly, D is a function of p: $D = d(p)$.

■ **EXERCISE 14** The demand function d satisfies $d'(p) = c\,d(p)$, with $c =$

(a) $0.2b$

(b) -0.2

(c) $-0.2b$

(d) 0.2.

Remember that, for $p > 0$ and a small number Δp, the following approximation holds:

$$d(p + \Delta p) \approx d(p) + d'(p)\,\Delta p$$

■ **EXERCISE 15** Suppose that the price of gasoline rises by 0.1 euro. Using the approximation above, we find that demand decreases by around ...

(a) 2 %

(b) 0.2 %

(c) 0.2 units

(d) 0.02 units.

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■ **EXERCISE 16** The price elasticity of demand is given by

(a) -0.2

(b) $-0.2D$

(c) $-0.2\,p$

(d) None of the options (a), (b) or (c) is correct.

Suppose an oil cartel controls the entire supply of gasoline and can price it as it wishes. Its revenues R are a function of this price p: $R = r(p)$.

■ **EXERCISE 17** The derivative r' of the revenue function r is zero if

(a) $p = 2$

(b) $p = 3$

(c) $p = 4$

(d) $p = 5$.

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